



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)

Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

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SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PRESENTS

ELECTROCOMM



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ABOUT THE DEPARTMENT

Welcome to the Department of Electronics & Communication Engineering (ECE). The Department of ECE at Kalasalingam Academy of Research and Education (KARE) was established in the year 1994. It is our great pleasure to announce that the department has recently attained accreditation for 6 years by the National Board of Accreditation (NBA) in 2022 under the Tier-I category. The department was also accredited by NBA for 3 years under the tier-I category in 2003, 2015, and 2018, respectively, and accredited by ABET for 6 years in 2020. The Department of ECE at KARE is among one of the oldest ones and offers B.Tech at UG level, M.Tech. (VLSI Design) at PG level, and Ph.D. in the Doctoral degree Programme. The Department offers an industrial Oriented B.Tech program in four streams including VLSI Design and Verification, Embedded Systems and IoT, AI for Cyber Security Professionals, and Microchip Physical Design under industry-academia collaboration with Nanochip Solutions, Bengaluru, India. The department has established branches, chapters, and societies of recognized professional bodies such as IEEE, IEEE-EDS, IETE, and ISTE. The department has state-of-the-art laboratory facilities enriched with the latest software viz. MATLAB, XILINX, Keil, TCAD, Synopsis, Optsim & LabVIEW, and a team of experienced & well-qualified faculty. The department offers skill and value-added certification courses on BEC, CLAD, SAP, testing of Electronic Equipment, simulation using PSPICE, Reconfiguration Design with HDL, OPTISIM simulation, Fundamentals of Android Applications, Electronic circuit design using Tina, Android-based IoT application, Photonic system design, Desktop GUI Application with JAVA, and Design of user Interface using Android and System Application & Products. The department has a good number of courses offered through online MOOC Platform (EDX, Udemy & NPTEL) and Foreign Academic Exchange with international partnering Institutions such as Asian University and Airlangga University. The department often organizes Faculty Development Programs and a few of the recently conducted FDPs include the sponsored event on the Future perspective of semiconductor devices, Artificial Intelligence & Deep Learning, Cyber-Physical Systems, deep learning application, Statistical and Machine learning, ASIC Design flow, Embedded IoT and the MOOC courses.

Department of Electronics and Communication Engineering

INSTITUTE VISION:

To be a University of Excellence of International Repute in Education and Research.



INSTITUTE MISSION:



To provide a scholarly teaching-learning ambience which results in creating graduates equipped with skills and acumen to solve real-life problems.
To promote research and create knowledge for human welfare, rural and societal development.
To nurture entrepreneurial ambition, industrial and societal connect by creating an environment through which innovators and leaders emerge.

DEPARTMENT VISION:

To strive towards excellence in Electronics and Communication engineering through teaching, experiential learning, quality research and scholarship, while adhering to ethical and societal requirements



DEPARTMENT MISSION:

To help students achieve their goals by recognizing, identifying, and fostering their unique strengths through quality education and cutting-edge research training.
To imbibe ability in the students to solve real life problems as per need of the society through nurturing their skills, creative thinking, and research acumen.
To create an elite workforce, without compromise in the ethics and societal values.

Program Educational Objectives (PEOs) (After June 2021)

PEO1: Technical Proficiency

Succeed as creative, productive, and valued engineers in their career by applying the technical knowledge and skills gained.

PEO2: Professional Growth

Continue to develop professionally through life-long learning, higher education, and research expertise to meet the expectations of their organisation and the society.

PEO3: Management Skills

Exhibit the management qualities in a responsive, ethical, and innovative manner.

B.TECH ECE PROGRAMME SPECIFIC OUTCOMES

At the end of the programme, the students will be able to:

PSO1: Knowledge:

Apply the knowledge of engineering and basic sciences to design, develop, test components and systems for applications including Signal Processing, VLSI Design, Embedded, Communication Engineering and Networking

PSO2: Skills:

Solve the complex engineering problems, using latest techniques, tools, along with the needed skills with an understanding of societal, environmental, safety, legal and cultural impacts of the solution.

PSO3: Attitude:

Apply the contextual knowledge of engineering to function effectively as an individual or a leader in multidisciplinary environments

B.TECH ECE PROGRAMME OUTCOMES (R2o18)

At the end of the programme, the students will be able to:

PO1: Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation to the solution of complex engineering problems.

PO2: Problem analysis:

Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

B.TECH ECE PROGRAMME OUTCOMES (R2o18)

PO7: Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning:

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

TABLE OF CONTENT

● CREATOR'S NOTE

FROM THE HOD'S DESK	1
EDITORIAL DESK	2
MEET THE TEAM	3

● TECHNICAL

VIKRAM SARABHAI	4
FACULTY ARTICLES	7
STUDENTS ARTICLES	25
TECHNICAL STORIES	42
INSPIRING PERSONALITIES	45
FUN TIME	46
AI BOTS	50
USEFUL WEBSITES	51

● NON-TECHNICAL

ESSAY	55
STORY	62
PENCIL- ART	66
PHOTOGRAPHY	73

● ACCOMPLISHMENTS

FEEDBACK FROM OUTGOING STUDENTS	79
PLACEMENT'S GALLERY	80 81



Dear Readers

Welcome to the latest issue of ELECTROCOMM -Volume 15, Issue 29. It is with great pleasure and enthusiasm that we present to you a diverse collection of stories, insights, and inspiration curated for your reading pleasure.

In the fast-paced world we live in ELECTROCOMM aims to be your guide, offering a moment of respite and discovery.

As we flip through these pages together, you will encounter thought-provoking articles, captivating visuals, and interviews with individuals who are making waves in their respective fields. Our goal is to create a space for meaningful dialogue, celebration of accomplishments, and exploration of the ideas shaping our world.

A magazine is more than just a collection of words and images; it's a journey. It's an exploration of new concepts, a celebration of diversity, and a platform for voices that deserve to be heard.

We extend our deepest gratitude to our talented contributors, whose creativity and dedication have brought this magazine to life. We also want to thank you, our readers, for your continued support and for being an essential part of the ELECTROCOMM community.

As you delve into the pages ahead, we encourage you to immerse yourself fully in the stories, connect with the ideas, and share your thoughts with us. Your feedback is invaluable and helps us grow.

Dr. A. Muthukumar
HOD, ECE



Dear Readers

Warm Greetings to all!!!

It gives me immense joy and satisfaction to pen in the 29th Issue, of our Department Magazine 'ELECTROCOMM' on this occasion as Editor. Magazines are the perfect platform that develops one's creativity. The biggest blocks to creativity is the inability to allow a thought or insight to have the space to become a possibility. Some of the inspired idea are the ones that start as crazy concepts. By allowing them to possibility, they may blossom into something usable. The Magazine aims to encourage creativity among young minds and inspire self-motivation. This issue includes articles, poems, sketches, photographs and technical, non- technical writeups and stories.

In each one, the contributors demonstrated their best aspects of their personalities through self-expression.

I would like to thank each and every contributor for their active participation in making the Magazine to float in our hands.

As an Editor, I hope you too enjoy reading this Magazine as much as we have enjoyed creating it.

A handwritten signature in black ink, which appears to read 'Dr. Abha'.

Dr. Abha Dargar
Asst. Prof, ECE

◀ MEET THE TEAM ▶

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Vikram
Sarabhai

Dr. Vikram Sarabhai

In this article, we will discuss the biography of Vibram Ambalal Sarabhai Jain who was an Indian physicist and astronomer who initiated space research and helped to develop nuclear power in India.

Vikram Sarabhai, born on August 12, 1919, in Ahmedabad, India, was a visionary scientist and the architect of India's space program. Hailing from a distinguished family, he displayed exceptional intellectual abilities from a young age. His academic journey took him to the United States, where he earned a doctorate in physics from the University of Cambridge in 1947. It was during this time that Sarabhai's interest in space science began to flourish.

Returning to India, Sarabhai dedicated himself to scientific research, particularly in cosmic-ray physics. In 1947, he founded the Physical Research Laboratory (PRL) in Ahmedabad, laying the groundwork for advancements in space and atmospheric sciences. Beyond traditional scientific pursuits, Sarabhai was a strong advocate for applying space technology to address national development challenges.

In 1962, Sarabhai established the Indian National Committee for Space Research (INCOSPAR), a precursor to the Indian Space Research Organisation (ISRO). His visionary goal was not just space exploration but leveraging space technology for societal needs and economic growth.

Under Sarabhai's leadership, India achieved a historic milestone with the successful launch of its first satellite, Aryabhata, in 1975. This achievement marked the beginning of India's prowess in space technology and exploration. Unfortunately, Sarabhai's life was cut short in 1971, leaving a void in the burgeoning Indian space program.

Vikram Sarabhai's impact extended beyond the realm of space science. Recognizing the importance of management education, he played a crucial role in establishing the Indian Institute of Management Ahmedabad (IIMA) in 1961. Additionally, he initiated the Community Science Centre in Ahmedabad, emphasizing the importance of promoting scientific knowledge among the public.

Sarabhai's holistic approach to development spanned education, science, and industry. His influence on policy formulation and institutional building significantly shaped India's scientific landscape. His outstanding contributions earned him numerous awards and honors both nationally and internationally.

The Vikram Sarabhai Space Centre (VSSC) in Thiruvananthapuram, Kerala, stands as a testament to his enduring legacy. This center, dedicated to space research and rocketry, embodies Sarabhai's vision of advancing space exploration in India.



Dr. Vikram Sarabhai Discoveries

Vikram Sarabhai helped in setting up many institutions all around the country and here are some of the well-known institutions established by Dr. Vikram Sarabhai.

- In 1947, Vikram Sarabhai established the Physical Research Laboratory (PRL) in Ahmedabad. PRL is a National Research Institute for space and allied sciences.
- Indian Institute of Management (IIM), Ahmedabad founded on 11 December 1961 is considered to be the best institute of management in the country.
- Uranium Corporation of India Limited (UCIL), Jaduguda, Bihar was founded in 1967, under the department of atomic energy.
- Vikram A. Sarabhai Community Science Centre (VASCSC) or the Community science centre was established in 1960 at Ahmedabad. VASCSC is working towards popularizing science and mathematics education among students, teachers, and the public. Its main objective is to improve and find innovative methods of scientific education.
- Darpan Academy for Performing Arts, Ahmedabad was founded in 1949 along with his wife and now directed by his daughter Mallika Sarabhai for the last three decades.
- Faster Breeder Test Reactor (FBTR), Kalpakkam was founded in 1985 and it is the testbed for fast fuel reactors and materials.
- Electronics Corporation of India Limited (ECIL), Hyderabad was founded in 1967 to create a strong indigenous base in electronics.
- Vikram Sarabhai Space Centre, Thiruvananthapuram founded on 21 November 1963, is a major space research centre of the ISRO which mainly focuses on rocket and space vehicles for the Indian satellite program.
- Space Applications Centre (SAC), Ahmedabad was founded in 1972. The Space Applications Centre has played an important role in realizing the vision and the mission of ISRO.

- Variable Energy Cyclotron Project or the VECC is located in Calcutta and was founded in 1972. VECC performs research in basic and applied nuclear sciences and the development of the nuclear particle accelerator.

Beyond his scientific achievements, Sarabhai's philosophy emphasized the use of technology for societal benefit. He envisioned space applications in telecommunications, weather forecasting, and resource management. This perspective laid the groundwork for ISRO's multifaceted mission to improve the quality of life in India.

Vikram Sarabhai's life and work are commemorated annually on National Science Day in India, observed on February 28th, the anniversary of his discovery of the Raman Effect. His legacy continues to inspire generations of scientists and space enthusiasts, underscoring the transformative power of vision, dedication, and the pursuit of knowledge.



"I also believe that a person who does not have respect for time, and does not have a sense of timing, can achieve little."

A large, light cream-colored circle with a rough, hand-drawn edge is centered on the page. It is surrounded by several dark brown, elongated, teardrop-shaped rays of varying lengths, creating a sunburst effect. The background is a solid light blue color.

Faculty Articles

THE HUMAN EAR RECOGNITION BASED ON PHASE-BASED MATCHING ALGORITHM

ABSTRACT: In this modern era, biometric play a vital role in authentication process for an individual identification. Broadly biometrics has been classified into anatomical and behavioral characteristics. Among the many biometrics, this paper focus on new emerging biometric named as Ear biometrics. In this paper, Phase-only Correlation (POC) and Band-limited Phase Only Correlation (BLPOC) has been proposed for ear recognition. The phase information has been obtained by calculating similarity between input and query ear image using 2D-Discrete Fourier Transform (DFT) and auto correlation function. Finally the phase-based on image matching have being success implemented for human ear recognition endeavor. The experimental resultant effect of proposed algorithm has been performed using IIT Delhi ear database.

1. INTRODUCTION

In impressive of last few years, Human Ear recognition has been becoming a very attractive in biometric authentication. The important reasons behind human ear biometric over other biometric modalities are smaller in size, very stable shape has proven by clinical observation (Rutty, Abbas, & Crossling, 2005), uniform colors, and does not affected by any expression like face or changes in age or position or rotation (Malathy, Annapurani, & Sadiq, 2013). The ear is a passive biometric as it can easily be captured from far distance, even if the person is not fully cooperative. Hence ear recognition is much interest for the researchers dealing with numerous applications in commercial and governmental, such as law enforcement, security systems and forensics (Jain, Ross, & Pankanti, 2006). Therefore, ear biometric is a deserved preference for it providing a fine avail between cost and accuracy.

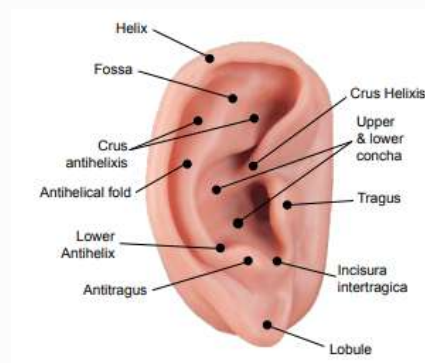


Figure 1. Ear Image

The Figure 1 shows the ear image of an individual. It is stored in the IIT Delhi database and can be used for the task of identification through its unique characteristics. The human outer ears parts are formed by various terminologies include Helix, the Lobe, Antihelix, the tragus, concha along with an antitragus.

Recently, several matching algorithms have been adapted for the ear recognition process. Anyhow, these algorithms possess various limitations which are explained in this section. Primarily, the holistic matching method utilized from global features to extract the complete ear. Bustard et al. arrange the ear dataset by considering the planar surface. It was registered as homography transform designed of Scale Invariant Feature Transform from trails (Bustard & Nixon, 2010). This method was strong to conditions in the appearance of pose variation and occlusion.

Ultimately, the matching algorithms with local feature have been adopted for identification. So Nanni et al. determined to decrease the dimension by usual sub-window of feature extraction with a Gabor Filter along with LaplacianEigenmaps (Zhu, Jia, & Liu, 2009). They also used the Sequential Forward Floating Selection to sort out the finest matching feature with concord trained. However the function of Laplacian Eigen maps could drop datas extremely. Yazdanpanah et al. designed a region of covariance matrix through well organized plus strong ear caption, which is strong to clarification with fake difference (Miyazawa, Ito, Aoki, Kobayashi, & Katsumata, 2006). In the following year, Yuan et al. ear region has extracted to segment by conserving from neighborhood impact and the particular region has chosen by the recognition rate (Zhang, Zhang, & Zhang, 2009). This method could keep away from the occlusion problem other than shortcoming from multi fake problem

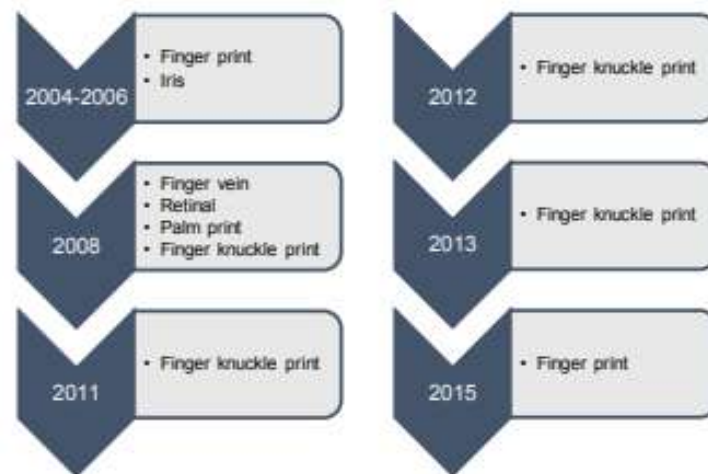


Figure 2. Different biometrics images used by POC and BLPOC

The POC and BLPOC matching algorithm play the major role in image processing or pattern matching. There are many benefits in using Phase based matching algorithm such as simple to construct score, easy to implement, finest for multiple test and close to real time performance. The Figure 2 shows different biometrics images are used by POC and BLPOC algorithm over the years for instance finger vein, finger print, iris, retinal, palm print as well as finger knuckle print except ear biometric. Therefore, this paper proposes human ear recognition based on phase-base matching algorithm.

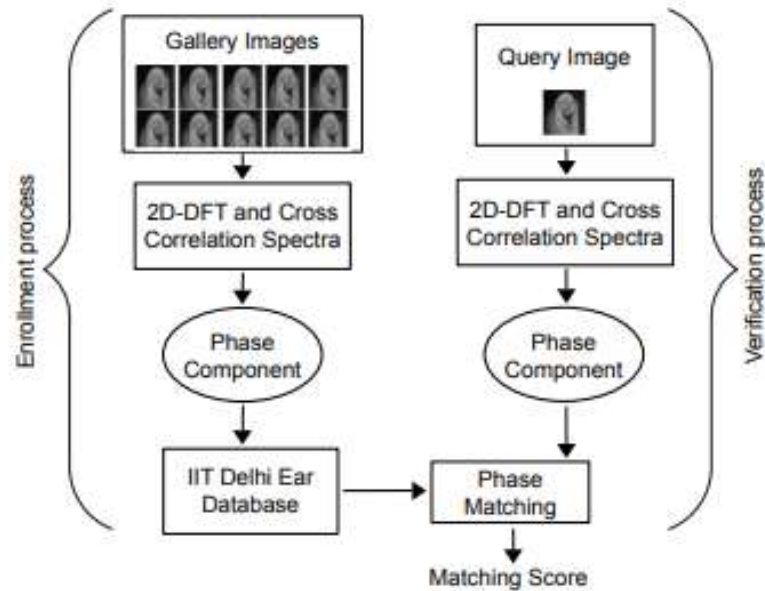


Figure 3. Proposed system

The Figure 3 shows the proposed system of this paper. This proposed system involves two processes namely (i) Enrollment process and (ii) Verification process. First, enrollment process carries out the process that involves feature extraction of POC and BLPOC technique. This feature extraction minimizes the ear images into mathematical phase information. This information will be stored in database to establish the authentication of ear images. At the same time, verification process also arrive the same process for extracting features of POC and BLPOC technique which generates the phase information. This feature extraction converts the ear images into mathematical phase information. These both phase information's are afforded by decision making subsidiary. The decision making subsidiary used to provide a final conclusion (i.e. genuine or imposter output). The output of decision making is a potential matching score by matching identities.

The remainders in this paper are structured as follows: The second section discusses about survey of the literature which gives brief discussion about existed POC and BLPOC. The third section outlines the proposed work and its implementation of Human Ear Recognition. The fourth section discusses about presents the experimental results of proposed system. The conclusion has been described in the section 5.

2. RELATED WORK

In this related work, discusses the detailed functions of phase-base on image matching algorithms (i.e. POC and BLPOC). The POC has been designed to find the correlation within two images (i.e. different images or similar images). The purpose of POC is to frame-up only phase feature and to isolate the magnitude feature of images for accuracy in image matching. During the trial time of POC with two similar images provides distinct sharp correlation peak and the correlation peak drops when two images are different (Takita, Muquit, Aoki, & Higuchi, 2004).

The experimental results of phase correlation demonstrate high robustness and accuracy in the pattern matching and the image registration (Ito, Nakajima, Kobayashi, Aoki, & Higuchi, 2004). This has been achieved with direct estimation in Fourier domain through 2D phase difference data set (Miyazawa et al., 2006). This kind of POC correlation is more convenient and more profitable than other conventional correlation. The most remarkable advantages of POC are it provides better discrimination for similar images and high accuracy translational of displacement among the illustrations (Nanni & Lumini, 2007). But for most part of phase function algorithm cannot carry the nonlinear deformation of illustrations (Rutty et al., 2005). The BLPOC is very crucial method for correlation in optimum band limit for phase based recognition. In POC function, every frequency components are concerned. Though, high frequency components tend to highlight and it probably prone to noise. Hence, worthless high frequency components are eliminated by setting a band-limited while calculating the cross-power spectrum (Yazdanpanah & Faez, 2010). Also, in order to carry the nonlinear deformation, it employs BLPOC matching algorithm. The BLPOC be sufficient to estimate accurate overall correlation (Yuan, Wang, & Mu, 2010). The modified version of POC is BLPOC which is committed to evaluate match between images due to its better performance than the POC function (Zhang, Zhang, Zhang, & Guo, 2012). The proposed BLPOC perform (Zhang, Zhang, Zhang, & Zhu, 2011) committed to biometric recognition tasks. Several authors attempted different modalities of biometric recognition by using POC and BLPOC are outlined in the given Table 1.

Table 1. Outline about Phase-Based Function matching algorithm in various biometrics.

Author	Feature	Trait (Images)
(Takita <i>et al.</i> , 2004).	POC and BLPOC	Finger Print
(Ito <i>et al.</i> , 2004).	POC and BLPOC	Iris
(Miyazawa, Ito, Aoki, Kobayashi, & Nakajima, 2005).	POC and BLPOC	Iris
(Zhu <i>et al.</i> , 2009).	POC and BLPOC	Palm print
(Mahri, Suandi, & Rosdi, 2010).	POC and BLPOC	Finger Vein
(Miyazawa <i>et al.</i> , 2006).	POC and BLPOC	Retinal
(Yuan <i>et al.</i> , 2010).	POC and BLPOC	FKP
(Zhang <i>et al.</i> , 2011).	POC and BLPOC	FKP
(Zhang <i>et al.</i> , 2012).	POC and BLPOC	FKP
(Aoyama, Ito, & Aoki, 2014).	POC and BLPOC	FKP

Human Ear Recognition Using Phase-Based Function Matching.

3. METHOD

In this section, POC and BLPOC are discussed and derived in detail.

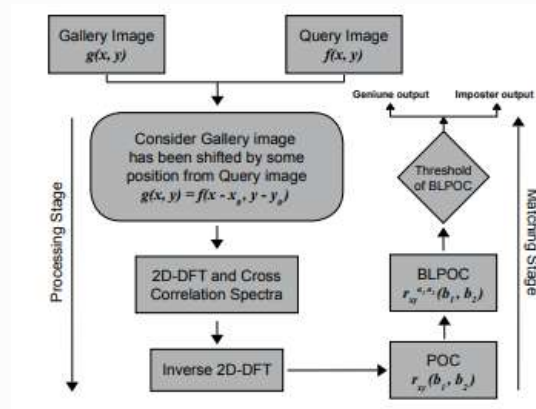


Figure 4. Overall structural outline of the proposed design of this paper

The Figure 4 shows the overall structural outline of proposed design of this paper. This method is discussed in next section 3.1 and 3.2 in detail.

5. CONCLUSION

This paper intends a human ear identification using phase based function matching algorithm. The proposed BLPOC algorithm of Human Ear Recognition makes it possible to align ear images, correctly evaluated similarity between them and obtained the reliable matching score. The experimental results reveal that the intended method has achieved the well again Human Ear recognition performance and robustness over other previous methods. Extensively experiments were tested on the Ear database IIT Delhi. The proposed method of human ear recognition has performed the finest verification results on the Ear database IIT Delhi, with the equal error rate 0.86%.



Dr. A. Muthukumar
HOD, ECE

A NOVEL ZERNIKE MOMENT BASED REAL-TIME HEAD POSE AND GAZE ESTIMATION FRAMEWORK FOR ACCURACY- SENSITIVE APPLICATIONS

Background- Real-time, robust head pose estimation algorithms have the potential to greatly advance in the fields of human computer and human-robot interaction. Head pose Estimation is used in the Driver's Assistance System. The Driver's Assistance System simultaneously analyzes the driver's head pose to road hazards. The main aim is prevention the road accidents due to inattentiveness behavior of the driver such as fatigue, drowsiness of the driver and distraction. The proposed system defines efficient vision based system to detect the person's direction of attention by estimating head pose and gaze direction automatically in order to improve the road safety. The researcher's implemented different approaches that are generally on image-based comparison generate the mapping functions, template matching, engineered facial feature points (both eyes, tip of the nose and mouth). In this paper, a hybrid scheme based on Zernike moments and Linear discriminant algorithm is implemented by considering the head poses and gaze direction information to obtain the enhance estimation results. Finally, the Experiment results on publicly available pose datasets or real time images shows that the proposed algorithm achieves accurate results and estimates the pose and gaze directions with faster speed and higher robustness even in large range of pose changes and invariant to the illumination conditions. The experimental results are compared with the Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA) to estimate Head Pose.

Introduction: The Head pose estimation calculates the spatial location of the human Head along with the position or rotation angle of the Head (front or side poses) in an image. Gaze estimation measures an eyeball direction to predict the human attention level and better understand human activities. Head pose always coincides with the gaze directions to know the focus of attention. It also successfully provides robust and accurate perception to interact with humans. Head pose Estimation is mainly used for analyzing complex meaningful gestures such as Head nodding or pointing gestures. Gaze direction is achieved by capturing and analyzing the appearance of the human's eyes images. The eyes are the windows of the brain, and most of the percentage, sensory information (contains the status of the brain information) is received by the eyes from brain to perform tasks.

The overall flowchart of the proposed automatic head poses and gaze direction estimation system is shown in figure 1. The proposed approach consists of two phases Training phase and Testing Phase. Preprocessing, Feature Extraction, dimensionality Reduction are common for both the train and test phases. Along with these steps, train phase consists of train model. The test image is passed as input to testing phase. Head Pose and Gaze Direction Estimation is recognized using Minimum Distance metric classifier.

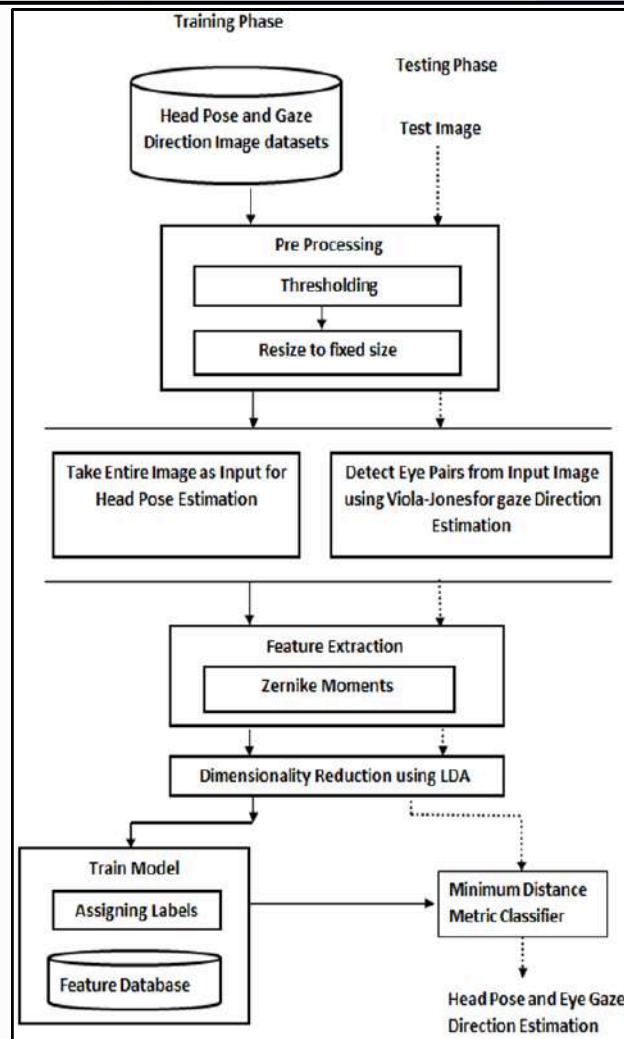


Figure 1: Architecture of Head Pose and Gaze Direction Estimation system

The performance is evaluated by taking the images from the public databases or real time images. It estimates the head pose and gaze direction with an accuracy of 85% using LDA. The system is unable to estimate correct results due to physical factors of the camera (low resolution and blurring) conditions. But most of the real time applications like driver assistance systems or interactive system need accuracy more than 90%. In future work, need to develop an approach to improve the performance of the system which is invariant to blurring conditions.



Dr. V. Hima Deepthi,
Professor, ECE

NOVEL TRI-ACTIVE LAYER CHANNEL AMORPHOUS-IGZO THIN FILM TRANSISTOR FOR NEXT-GENERATION DISPLAY APPLICATIONS

Among high field mobility materials, Amorphous-Indium-Gallium-Zinc-Oxide have been prominent for their use in Thin Film Transistors (TFTs). In this letter, the authors present a novel structure of TFT with Tri-Active Layer (TAL) channel consisting of amorphous In-Ga-Zn-O (α -IGZO) and In-Sn-O (ITO) materials. Further, the simulation-based analysis of the proposed device and characteristics with comparison to the single and double active layer channel structures have been reported. This work also analyzes the effect of front-back channel thickness-ratios in double-active layer channel TFTs. The proposed TAL channel TFT describes improved switching parameters such as lower threshold voltage (V_{TH}) -4.24 V, high field-effect mobility (μ_{FE}) 38.4 cm²/V-s, moderate Subthreshold Swing (SS) 0.398 mV/decade, and reasonably high ON to OFF current ratio (I_{ON}/I_{OFF}) 0.18 $\times$ 10¹². The electrical characteristics persuade the significance of the proposed structure in next-generation flat-panel displays.

Introduction: Transparent amorphous conducting oxide thin films have been considered for various application areas such as switching element in the backplane of Organic Light Emitting Diode (OLED) displays, ultra-high definition Liquid Crystal Displays (LCDs), electrophoretic displays, opto-electrical devices, gas sensors, solar cells, and large area electronics. Presently, Zinc-based Transparent Conducting Oxides (TCOs), such as GZO, [6], IGZO, [7], SGZO, ZnO, ZIO, and IZTO are of great importance in the field of TFTs. In this research work, we have designed a new TFT of multi-stack Tri-Active Layer (TAL) channel structure and obtain the electrical parameters from their output and transfer characteristics analysis. Also, the effect of thickness variation of double-active layer channel to the TFT performance has been realized and finally presents a performance comparison of Single, Double, and Tri-Active Layer channel based TFTs.

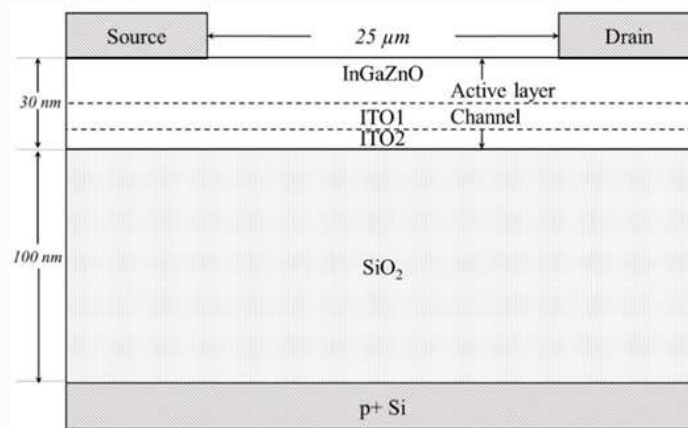


Figure 1: Schematic of the proposed device (bottom-gate top-contact)

Primarily, the active channel has 27 nm-thick α -IGZO layer in the TAL structure. The single, double and tri-active layer channel of the TFT which are distinguished particularly in terms of the density of states of the active channel, has been analyzed using the device simulator. The simulation is backed with parameters such as bandgap 3.2 eV for both ITO and α -IGZO. In the AMOLEDs Pixel circuit, α -IGZO TFTs have been prominent due to its notable advantages i.e. high mobility, stability and low temperature processing. AMOLED pixel driver circuit using Double-Gate TFTs have been proposed to overcome electrical instability issues of Single Gate TFTs which in turn causes threshold voltage shift when subjected to continuous gate bias-stress.

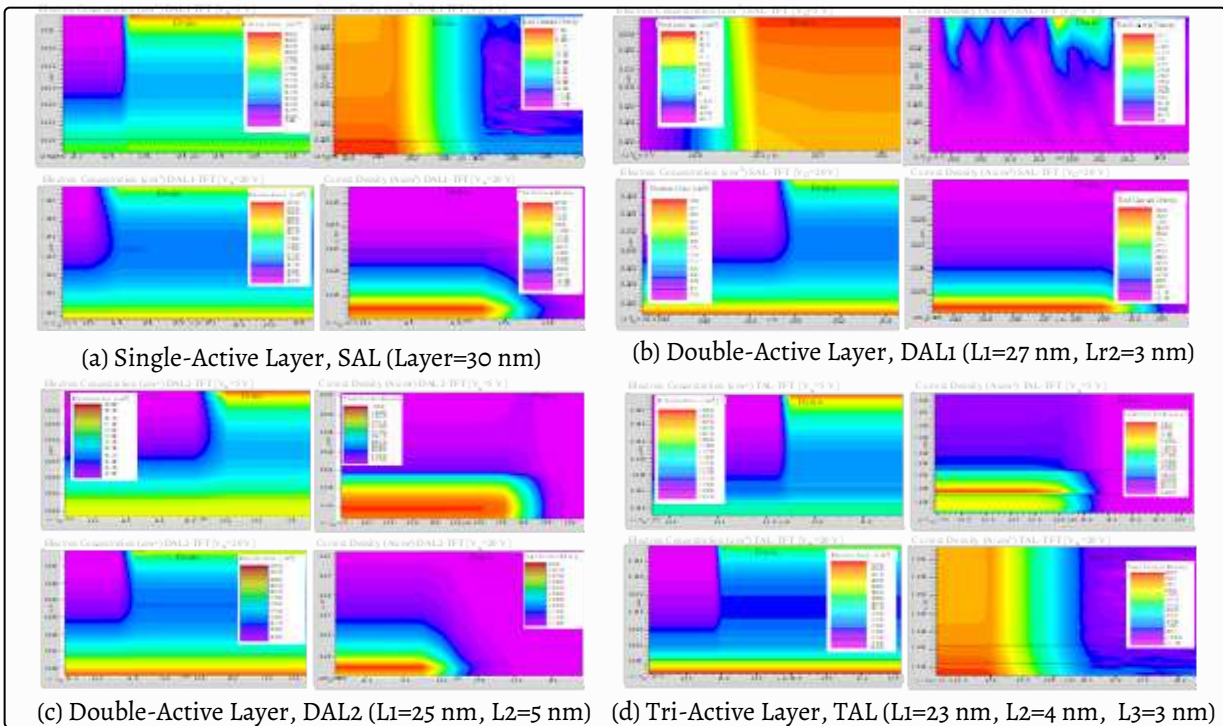


Figure 2: Simulation of the active layer in terms of electron concentration and current density at $V_G=5$ V and $V_G=20$ V.

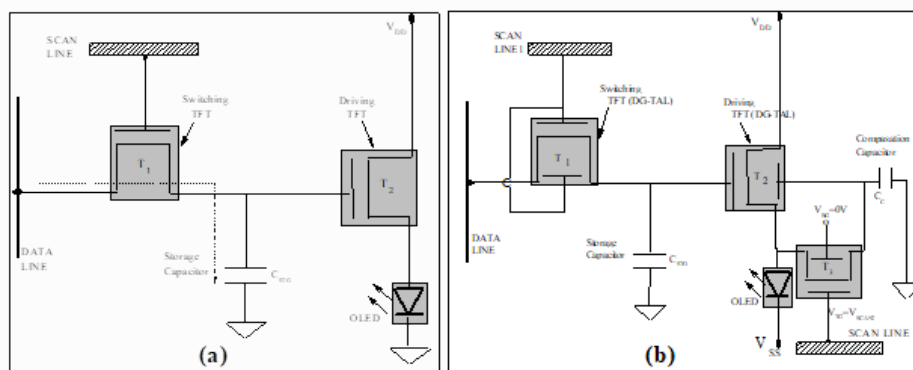


Figure 3: AMOLED pixel circuit (a) conventional TFT (b) Proposed (DG)TAL α -IGZO based TFT.

A conventional pixel circuit using Double gate is comprised of two TFTs and a capacitor as shown in Fig. 3 (a) , which has one T1 (Switch-TFT) and the other is T2 (Drive-TFT). T1 switches ON and OFF in program interval and capacitor gets charged up to data voltage level. In its ON state, T2 supplies constant current for driving OLED from the capacitor (CSTO) voltage. If the capacitor is not fully charged with in the program interval, it leads to display error. Hence the switching TFT must have higher ION/IOFF ratio for faster switching. As the designed DG-TAL TFT in offset structure has shown low OFF current along with superior electrical parameters, it will certainly be improving the charging performance of capacitor in the pixel circuit. The AMOLED pixel circuit utilizing DG-TAL TFT has been proposed as shown in Fig. 3(b) with two DG-TAL TFTs and one capacitor as switching, driving, and storage element, respectively. The line added for attaining good control of threshold voltage and compensation in the proposed AMOLED pixel circuit.



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Associative Professor, ECE

MOS2: THE NANOMATERIAL TRANSFORMING TECHNOLOGY'S FRONTIERS

Background: Unveiling the Potential of a Molecular Marvel

In the rapidly evolving world of nanotechnology, molybdenum disulfide (MoS₂) has emerged as a groundbreaking material that promises to reshape multiple technological domains. Unlike traditional materials, MoS₂ stands out with its unique two-dimensional (2D) structure, offering unprecedented capabilities that extend far beyond conventional technological limitations.

At its core, MoS₂ is a transition metal dichalcogenide with extraordinary properties. Its atomic-thin structure comprises a layer of molybdenum atoms sandwiched between two layers of sulfur atoms, creating a molecular architecture that enables remarkable versatility. What makes MoS₂ truly exceptional is its ability to adapt and perform across diverse technological applications, from electronics and sensors to biomedical innovations.

The material's most fascinating characteristic is its tunable bandgap, which can be modified from 1.29 to 1.9 electron volts by simply altering the number of layers. This quantum-level flexibility allows researchers to customize MoS₂ for specific technological requirements, making it a chameleon-like substance that can be precisely engineered for different purposes.

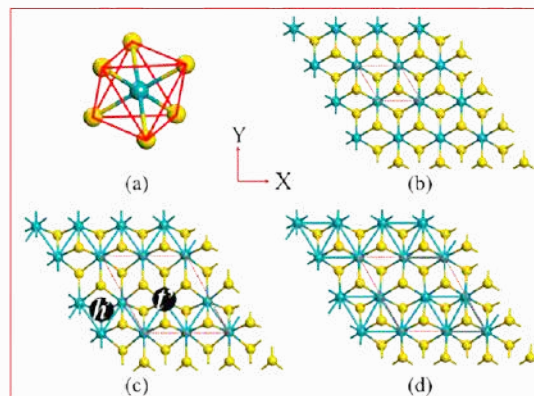


Fig. 1. (a) Structural unit of octahedral coordinated 2D MoS₂. Geometric structures of (b) T-MoS₂ (c) ZT-MoS₂ and (d) D-MoS₂

Introduction: MoS₂ is not just another nanomaterial—it's a technological Swiss Army knife with potential applications that span multiple critical sectors. Its remarkable properties have caught the attention of researchers worldwide, who are exploring its potential in areas ranging from ultra-low-leakage memory devices to sophisticated biosensors and renewable energy technologies.

In the realm of electronics, MoS₂ is revolutionizing transistor design. Its atomic-thin structure allows for unprecedented scalability, offering solutions to the limitations faced by traditional silicon-based technologies. Researchers have demonstrated MoS₂ transistors with remarkably low leakage, a critical parameter for developing next-generation electronic devices.

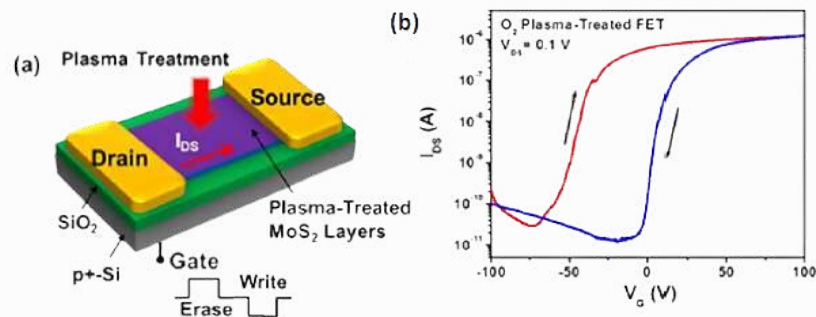


Fig. 2. a) The representation of a MoS2 Field Effect Transistor treated with plasma b) The curve of transfer characteristic (I_{ds} vs. V_G).

The material's sensing capabilities are equally impressive. MoS2-based sensors can detect everything from dangerous gases like NO2 and NH3 to complex biomolecules associated with serious diseases. In one breakthrough study, researchers developed a MoS2 field-effect transistor capable of detecting Down syndrome-related DNA sequences with incredible sensitivity—achieving detection limits as low as 100 attoMolar.

Energy applications represent another frontier where MoS2 is making significant strides. The material has shown exceptional promise in solar cell technology, serving multiple roles such as hole transport layers, electron transport layers, and buffer zones. In hydrogen evolution reactions, specific phases of MoS2 have demonstrated remarkable catalytic properties, offering potential solutions for clean energy generation.

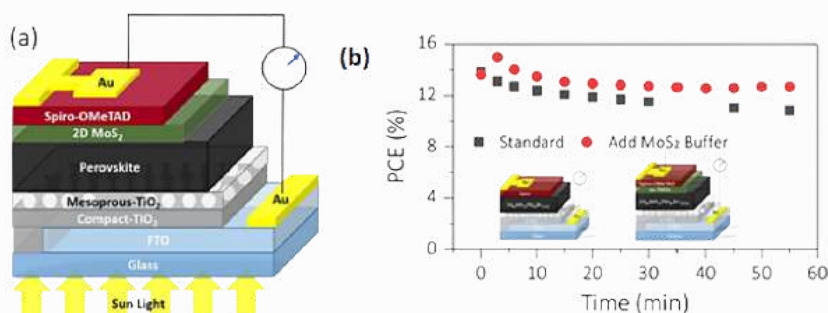


Fig. 3. a) Perovskite solar cell schematic with 2D MoS2 as a buffer layer; b) power conversion efficiency of standard solar cell and modified cell with MoS2 layer

Perhaps most excitingly, MoS2 is making substantial contributions to biomedical technologies. Its high biocompatibility and unique optical properties enable advanced biosensing techniques. Researchers have utilized MoS2 to develop sensors capable of detecting various biomolecules, opening new pathways for early disease detection and personalized medicine.

Despite its immense potential, the journey of MoS2 from laboratory breakthrough to widespread technological implementation is not without challenges. Researchers are actively working to overcome current limitations, such as improving material consistency, enhancing manufacturing processes, and scaling up production.

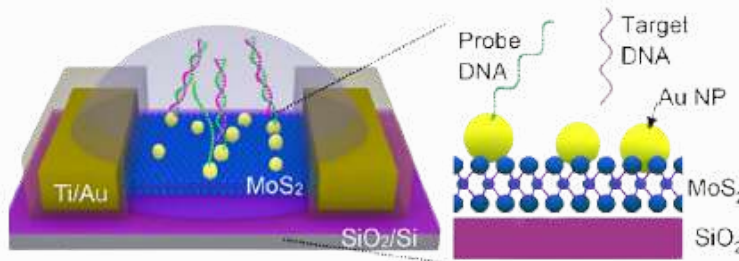


Fig. 4. The MoS₂ FET biosensor

The true magic of MoS₂ lies in its potential for hybrid structures—combining it with other two-dimensional materials like graphene could unlock even more revolutionary capabilities. As interdisciplinary research continues to push the boundaries of what's possible, MoS₂ stands as a testament to human innovation, bridging the gap between fundamental scientific discovery and transformative technological solutions. The coming decades promise to be an exciting era of exploration, with MoS₂ potentially playing a pivotal role in addressing some of our most pressing technological and environmental challenges.

As we stand on the cusp of a technological revolution, MoS₂ represents more than just a material it embodies the future of interdisciplinary scientific innovation. Its ability to bridge multiple technological domains suggests we are witnessing the early stages of a transformative era in nanotechnology.



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SCALING NEUROMORPHIC TECHNIQUES WITH 3D TECHNOLOGIES

Neurons generally have a tree-like structure comprising multiple dendritic arbors, which receive inputs, a cell body or soma that integrates these inputs, and an axon that transmits the neuron's output to others. When the input integrated at the soma exceeds a certain threshold, the neuron generates an electrical pulse, known as a spike. This spike travels along the axon, forming synaptic connections with the dendrites of subsequent neurons. In the brain, each neuron connects to approximately 10,000 others, and this extensive connectivity is fundamental to its cognitive abilities.

Neuromorphic hardware is designed to emulate the structure and function of neuronal networks. In the brain, extensive connectivity is achieved through physical links, such as axons, for each spiking pathway. However, in artificial systems, which typically rely on electronic substrates like complementary metal–oxide–semiconductor (CMOS) technology, it is not feasible to create adaptable connections like those in the brain, especially since these systems are built in two-dimensional layouts, unlike the brain's three-dimensional architecture. Consequently, connectivity in artificial systems is established through various communication channels and routing strategies, enabling flexible, reconfigurable, and virtual connections over a fixed wiring infrastructure.

In spike-based neuromorphic systems, communication relies on the timing and frequency of spikes—or events—generated by artificial neurons. The most widely adopted communication protocol in these systems is address event representation (AER). This protocol leverages the speed mismatch between the firing rate of artificial neurons and the signaling rate of CMOS wires, enabling time multiplexing to consolidate multiple firing events into a single communication channel.

These communication networks require on-chip memory to store connectivity information, or synapses, between computational units (neurons). However, as the size of on-chip networks increases, two significant challenges emerge: the need for substantial on-chip storage, which incurs high costs, and the excessive data movement within the network. This increased data transfer reduces bandwidth while raising latency and power consumption.

The issue of data movement can be solved with in-memory computing architectures, which co-locate memory and data processing¹. Large memory arrays are still required to store the vast amounts of connection data, and parasitics on the lines limit the amount of memory that can be used because they slow down the propagation of signals and compromise their integrity. Moreover, the complexity of the peripheral circuitry increases with the size of the array. Medium-sized fully connected neural network layers are best implemented using crossbar-based in-memory compute arrays. However, in order to transport data between several cores, large-scale systems must switch from single in-memory processing cores to multi-core systems, which reintroduces data movement issues.

Three-dimensional (3D) technology offers a promising solution (Fig. 1a) by enabling the vertical stacking of logic, memory, and sensing components through unconventional fabrication techniques. This approach can significantly enhance on-chip storage capacity, allowing the weights of large neural networks to be fully accommodated. It also reduces latency and power consumption due to shorter interconnections, improves bandwidth between stacked layers via multiple connections, and facilitates the integration of heterogeneous layers, each optimized for specific functions.

There are four primary 3D technologies: 3D NAND flash memory; 3D back-end-of-line (BEOL) integration of memory onto front-end-of-line (FEOL) logic; sequential 3D integration of multiple FEOL logic and BEOL memory layers; and 3D packaging, which vertically combines heterogeneous technologies (logic, memory, and sensing) across multiple dies. However, for these advancements to be successful, challenges related to hierarchical connectivity in neuromorphic chips must be effectively addressed. In this section discussed about 3D NAND flash memory; 3D back-end-of-line (BEOL)

3D NAND flash memory

Early efforts in developing 3D technologies for neuromorphic systems primarily centered on designing high capacity 3D memories to store network parameters like synaptic weights. Among various non-volatile memory options, 3D NAND flash emerged as a leading choice due to its commercial maturity and exceptional memory density, exceeding 10 Gb mm^{-2} (ref. 2).

However, this approach has its drawbacks. Notably, 3D NAND flash operates as standalone off-chip memory, necessitating the use of at least two distinct wafers: one for logic transistors and another for high-voltage transistors and 3D NAND cells. This separation poses an energy-efficiency challenge, as the cost of memory access rises with the physical distance between the memory and the processing unit, resulting in what is commonly referred to as the memory bottleneck.

3D BEOL integration of memory on FEOL logic

The second 3D technology focuses on incorporating non-volatile memristive devices—such as resistive random access memory (ReRAM), magnetic random-access memory (MRAM), phase-change memory (PCM), and ferroelectric random-access memory (FeRAM)—into the back end of line (BEOL), layered over advanced CMOS logic. These technologies are characterized by their fast switching and low programming voltage, making them well-suited for on-chip BEOL memories in advanced technology nodes. This approach effectively addresses the memory bottleneck posed by off-chip NAND flash. Additionally, they naturally enable an in-memory computing architecture, significantly reducing the need for frequent memory access.

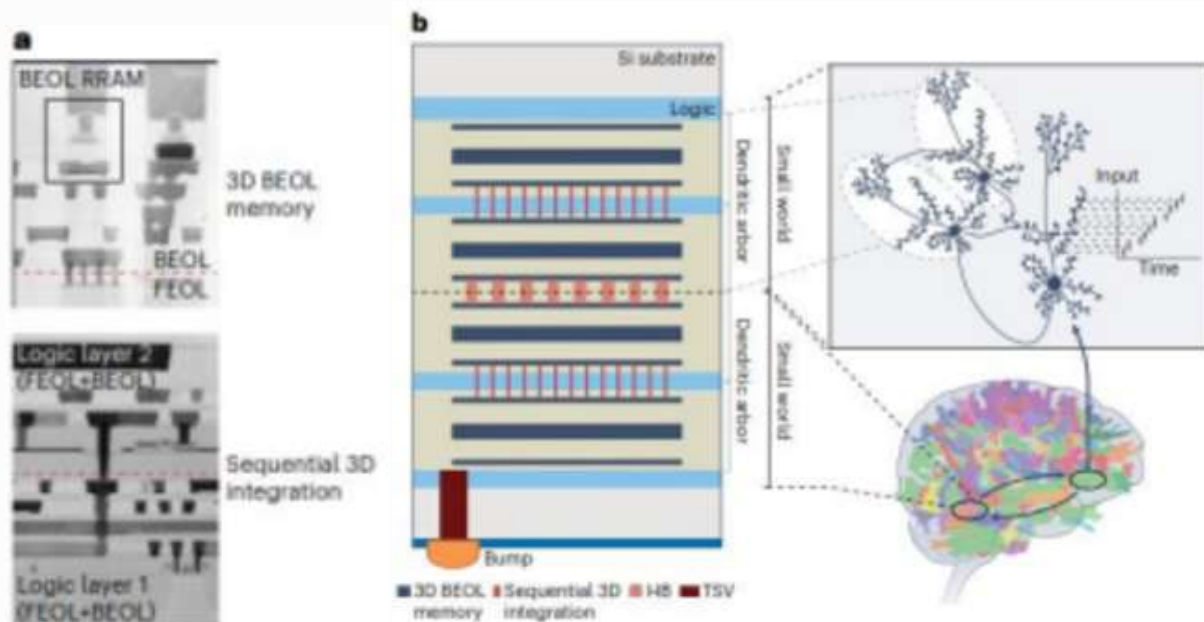


Fig. 1 Building 3D neuromorphic systems.

This architecture utilizes programmable memristors positioned at grid intersections, forming crossbar arrays that store a matrix of weights as memristor conductance. The input signal, represented as a voltage vector, is applied to the array. Using Ohm's law, the voltage inputs interact with the weight conductances to produce currents, which are subsequently summed in accordance with Kirchhoff's law. This process naturally executes the fundamental neural network operations of multiplication and accumulation.

To represent network parameters more efficiently, devices must exhibit multi-level conductance properties. This necessitates the inclusion of an access transistor alongside the memristors to enhance control over device programming. However, the size of the access transistor restricts the integration density of memristor arrays. To overcome this limitation and achieve higher memory density, multiple layers of back-end-of-line (BEOL) access transistors and memory are constructed sequentially.

For instance, two memory layers—each consisting of a ferroelectric capacitor combined with a stackable BEOL polycrystalline silicon transistor—have been successfully fabricated on top of silicon CMOS. Alternatively, monolithic integration methods have enabled stacking BEOL carbon nanotube transistors and resistive random access memory on silicon CMOS. This sequential integration approach offers greater flexibility, allowing the co integration of various types of memristors.

Implementing such systems with 3D nanotechnology-enabled architectures necessitates a shift away from traditional design methodologies. This transition requires a comprehensive approach, originally conceptualized by Misha Mahowald and Carver Mead in the 1990s, which integrates device and circuit design with system-level perspectives and insights from neuroscience.



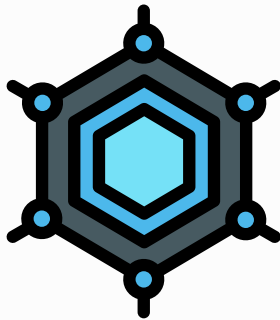
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NANO ELECTRONICS

Introduction

In the ever-evolving landscape of technology, the semiconductor industry has been the driving force behind the exponential growth of computing power and electronic devices. As we venture further into the 21st century, the need for smaller, faster, and more energy-efficient electronic components has become increasingly crucial. Enter nanoelectronics, a field that holds the promise of revolutionizing the way we design and manufacture electronic devices.



Nanoelectronics is a branch of nanotechnology that deals with the development and application of electronic components and devices at the nanoscale. This scale typically involves structures and materials with dimensions on the order of nanometers, where a nanometer is one-billionth of a meter. To put this in perspective, a typical human hair is about 50,000 to 100,000 nanometers wide. Here are some key details about nanoelectronics:

Nanometer Scale:

Nano electronic devices are characterized by their extremely small size, often with critical dimensions less than 100 nanometers. These tiny dimensions allow for the manipulation of individual atoms and molecules to control electronic properties.

Quantum Effects:

At the nanoscale, quantum mechanical effects become prominent and can significantly influence the behavior of electronic devices. Quantum dots, for example, exhibit discrete energy levels and quantum tunneling, which are exploited in various applications.

Nanomaterials:

Nanoelectronics often involve the use of novel nanomaterials, such as carbon nanotubes, graphene, nanowires, and quantum dots. These materials have unique electronic properties and can be used to create innovative devices.

Transistors:

The transistor is a fundamental building block of electronic circuits, and nanoelectronics has seen the development of nanoscale transistors, such as single-electron transistors (SETs) and tunnel field-effect transistors (TFETs). These transistors operate at low power and offer high performance.

Nanoelectronics encompasses a wide range of topics, including the fabrication of nanoscale transistors, the exploration of novel materials, and the development of new circuit designs to harness the unique properties exhibited by matter at the nanoscale.



The general equation linking the current I to applied voltage V is : $I = I_0 \left[\exp \left(\frac{eV}{nkT} \right) - 1 \right]$

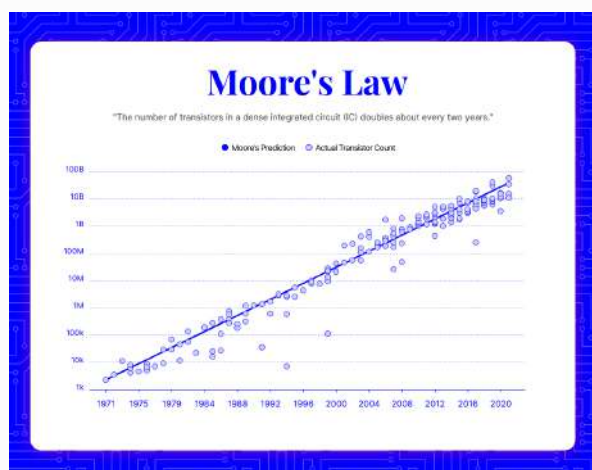
Need of Nanoelectronics

The demand for nanoelectronics arises from several key factors:

1. Moore's Law and Miniaturization
2. Energy Efficiency
3. Novel Materials

1. Moore's Law and Miniaturization:

For decades, the semiconductor industry has been guided by Moore's Law, which observes that the number of transistors on a microchip tends to double approximately every two years. This relentless pursuit of miniaturization has led to the remarkable growth of computing power, but as we approach physical limits, traditional silicon-based electronics face challenges related to power consumption and heat dissipation. Nanoelectronics offers a pathway to continue the trend of miniaturization beyond what was once thought possible.



2. Energy Efficiency:

As electronic devices become more integrated into our daily lives, energy efficiency is paramount. Nano electronic components, due to their tiny size and unique properties, have the potential to drastically reduce energy consumption in electronic devices. This is especially important in applications like mobile devices and IoT (Internet of Things) devices, where battery life is a critical factor.

3. Novel Materials:

Nanoelectronics introduces the use of novel materials such as carbon nanotubes, graphene, and quantum dots, which possess exceptional electrical and thermal properties. These materials enable the development of electronic components that are not only smaller but also more efficient and versatile. For instance, carbon nanotube transistors have the potential to replace traditional silicon transistors and offer superior performance.

Advantages of Nanoelectronics

- **Miniaturization:** Nanoelectronics enables the miniaturization of electronic components and devices to the nanoscale, which allows for the creation of smaller, more compact, and lightweight electronic systems. This is particularly valuable in portable devices and integrated circuits.
- **Higher Performance:** Nano electronic devices can exhibit higher performance in terms of speed, power efficiency, and computational power due to the unique properties of nanomaterials and quantum effects. This is crucial for applications like computing and data processing.
- **Lower Power Consumption:** Nano electronic components often operate at lower power levels, which can lead to reduced energy consumption and longer battery life in electronic devices.
- **Improved Sensing and Detection:** Nano-electronics enables the development of highly sensitive sensors and detectors that can detect minute changes in physical, chemical, or biological parameters. This is valuable in fields like healthcare and environmental monitoring.
- **Novel Materials:** Nanoelectronics relies on the use of novel nanomaterials, such as graphene and carbon nanotubes, which have unique electronic properties and can outperform traditional materials in certain applications.
- **Quantum Computing:** Nanoelectronics plays a crucial role in the development of quantum computing, which has the potential to solve complex problems far more efficiently than classical computers.

$$\text{MOSFET drain current : } I_D = K(V_{GS} - V_{TH})^2$$

Disadvantages of Nanoelectronics

- **Fabrication Challenges:** Building Nano electronic devices with precision and reliability can be extremely challenging. Advanced fabrication techniques, often requiring expensive equipment, are needed, which can increase production costs.
- **Quantum Effects:** While quantum effects can provide advantages, they can also introduce unpredictability and noise into Nano electronic systems, which can be problematic in some applications.
- **Heat Dissipation:** As devices become smaller and more powerful, managing heat becomes increasingly difficult. Heat dissipation issues can lead to overheating and reduced device lifespan.
- **Cost:** The development and production of Nano electronic components can be costly, and this expense may limit their widespread adoption, especially in consumer electronics.
- **Durability and Reliability:** Nano electronic devices may be more susceptible to environmental factors, such as radiation and contamination, which can impact their long-term reliability.
- **Ethical and Environmental Concerns:** The production and disposal of Nano electronic materials may raise ethical and environmental concerns, especially regarding the potential health and environmental impacts of nanomaterials.
- **Security Risks:** The small size of Nano electronic components can also pose security risks, as they may be vulnerable to tampering or counterfeiting.

Applications of Nanoelectronics

The impact of nanoelectronics is far-reaching and extends into various sectors:



N – channel MOSFET cut off : $V_{DS(Sat)} = V_{GS} - V_{TH}$

1. Computing:

Nanoelectronics is poised to redefine the computing landscape. Quantum computing, a subset of nanoelectronics, promises to solve complex problems that are practically unsolvable with classical computers. This could have profound implications for fields such as cryptography, drug discovery, and materials science.

2. Healthcare:

In the medical field, nanoelectronics enables the development of sophisticated diagnostic tools, targeted drug delivery systems, and even miniature implantable devices that can monitor and treat medical conditions from within the body.

3. Energy:

Nanoelectronics plays a crucial role in renewable energy technologies. For instance, the development of highly efficient, nanoscale solar cells and energy storage systems holds the potential to revolutionize how we generate and store energy.

4. Electronics Manufacturing:

The manufacturing of Nano electronic components demands cutting-edge fabrication techniques, which in turn drive advances in materials science and nanoscale engineering. This has a ripple effect on the broader electronics manufacturing industry.

5. Memory Devices:

Nanoelectronics has led to the development of non-volatile memory devices like Flash memory and Resistive Random-Access Memory (RRAM) with higher storage densities and faster access.

6. Sensors:

Nano electronic sensors are highly sensitive and can detect extremely small quantities of substances. They are used in applications like medical diagnostics, environmental monitoring, and security.

7. Quantum Computing:

Nanoelectronics plays a crucial role in the development of quantum computing, where qubits (quantum bits) are implemented using nanoscale components.

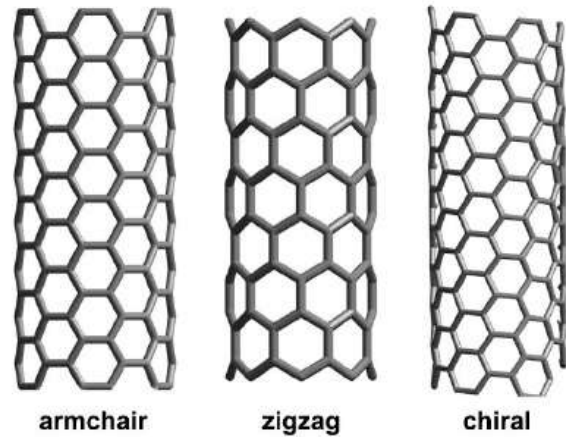
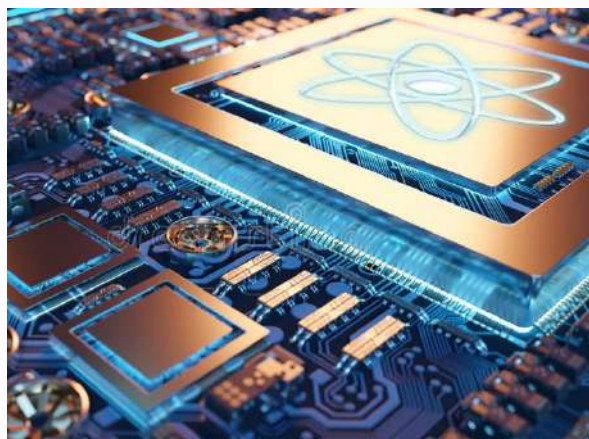
8. Energy Harvesting:

Nano electronic materials and devices are used in energy harvesting applications, such as nanogenerators that convert mechanical energy into electrical energy at the nanoscale.



Challenges and Ethical Considerations

While nanoelectronics holds great promise, it also presents several challenges and ethical considerations. These include issues related to environmental impact, potential job displacement, and privacy concerns, especially in the context of nanoscale surveillance technologies.



The Future of Nanoelectronics

Nanoelectronics is at the forefront of technological innovation, with researchers and engineers pushing the boundaries of what is possible at the nanoscale. As we move forward, collaborations between academia, industry, and government organizations will be crucial to address the challenges and harness the full potential of this exciting field. Nanoelectronics is not just about making electronics smaller-it's about creating a future where electronics are more efficient, powerful, and sustainable than ever before. The field of nanoelectronics continues to evolve. Researchers are exploring novel materials, like 2D materials, and new device concepts, such as spintronics and topological insulators, to push the boundaries of what is possible in electronic technology. In summary, nanoelectronics is a multidisciplinary field that combines nanotechnology and electronics to create advanced electronic devices and components with unique properties and capabilities. It has the potential to drive innovations in computing, sensing, energy, and various other technological domains.

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$$MOSFET \text{ saturation : } I_D = \frac{\mu C_{as}}{2} \cdot \frac{W}{L} (V_{GS} - V_T)^2$$

AUGMENTED REALITY

Introduction:

Augmented reality (AR) is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli and delivered via technology. It is a growing trend among companies involved in mobile computing and business applications in particular.

What Is Augmented Reality Used for?

Augmented reality either makes visual changes to a natural environment or enhances that environment through adding new information. It can be used for a variety of purposes, including gaming, product visualization, marketing campaigns, architecture and home design, education, and industrial manufacturing, among others.

What Are the Advantages of Augmented Reality?

Augmented reality is an interactive experience in which a real world environment is enhanced with computer-generated visual elements, sounds, and other stimuli. It can provide a user with a heightened, more immersive experience than they would experience otherwise that adds to the user's enjoyment or understanding. From a commercial perspective, augmented reality can increase brand awareness and boost sales.



Is Augmented Reality or Virtual Reality More Realistic?

Since augmented reality uses a real world environment in conjunction with computer-generated visual elements, it may be seen as more realistic than virtual reality, which is 100% computer-generated. However, advances in technology mean that even entirely virtual environments can seem fully realistic.

The main differences between augmented reality (AR) and virtual reality (VR) are:

- AR uses a real-world setting while VR is completely virtual.
- AR augments the real-world scene, whereas VR creates completely immersive virtual environments.
- AR enhances both the virtual and real world while VR only enhances a fictional reality.
- AR is 25% virtual and 75% real, while VR is 75% virtual and 25% real.
- AR can be accessed with a smartphone, while VR requires a headset device.

$$\text{Band gap variation} : E_G(T) = E_G(O) - \beta_0 T$$

Key Concepts and Features of Augmented Reality:

• Real-time Interaction:

AR applications provide real-time interaction with the physical environment. Users can see, hear, and sometimes even feel digital elements integrated seamlessly into their surroundings.

• Marker-Based vs. Marker less AR:

AR can be categorized as marker-based, where it relies on visual markers (such as QR codes) to trigger digital content, or marker less, which uses sensors like GPS, accelerometers, and computer vision to identify the user's environment.

• Applications:

AR has a wide range of applications across various industries. Some common uses include gaming (e.g., Pokémon GO), navigation (e.g., AR navigation apps), education (e.g., interactive textbooks), healthcare (e.g., medical simulations).

Hardware:

To experience AR, users often require specific hardware. This can range from smartphones and tablets for mobile AR to specialized AR glasses like Microsoft HoloLens or Google Glass.

• Computer Vision:

AR relies heavily on computer vision, a field of artificial intelligence (AI) that enables machines to interpret and understand the visual world.

This technology allows AR devices to recognize objects, locations, and gestures.



• Challenges:

AR faces challenges like ensuring accurate tracking, creating realistic 3D models, reducing latency, and optimizing battery life for mobile devices.

Additionally, privacy and ethical concerns related to AR data collection and use need to be addressed.

• Future Trends:

The future of AR holds the promise of more immersive and interactive experiences. Advancements in hardware, such as improved AR glasses, and innovations in software, like better spatial mapping and AI integration, are expected to drive AR's growth.

• AR in Business:

Many businesses are adopting AR for purposes like training, remote assistance, and virtual product demonstrations. It can enhance employee productivity and customer engagement.

• AR in Entertainment:

The entertainment industry has been at the forefront of AR adoption, with applications in gaming, live events, and interactive storytelling.

• Education and Training:

AR is increasingly being used in education and training, providing students and professionals with hands-on, immersive learning experiences.

$$\text{Fermil level : } f(E) = \frac{1}{1 + e^{(E-E_F)/KT}}$$

The Present State of Play in Augmented Reality:

Augmented reality is achieved through a variety of technological innovations; these can be implemented on their own or in conjunction with each other to create augmented reality.

They include:

General hardware components – the processor, the display, the sensors and input devices.

Typically a smartphone contains a processor, a display, accelerometers, GPS, camera, microphone etc. and contains all the hardware required to be an AR device.

Displays – while a monitor is perfectly capable of displaying AR data there are other systems such as optical projection systems, head-mounted displays, eyeglasses, contact lenses, the HUD (heads up display), virtual retinal displays, EyeTap (a device which changes the rays of light captured from the environment and substitutes them with computer generated ones), Spatial Augmented Reality (SAR – which uses ordinary projection techniques as a substitute for a display of any kind) and handheld displays.

Sensors and input devices include – GPS, gyroscopes, accelerometers, compasses, RFID, wireless sensors, touch recognition, speech recognition, eye tracking and peripherals.

Software – the majority of development for AR will be in developing further software to take advantage of the hardware capabilities. There is already an Augmented Reality Markup Language (ARML) which is being used to standardize XML grammar for virtual reality. There are several software development kits (SDK) which also offer simple environments for AR development.



The Future of Augmented Reality:

Jessica Lowry, a UX Designer, writing for the Next Web says that AR is the future of design and we tend to agree. Already mobile phones are such an integral part of our lives that they might as well be extensions of our bodies; as technology can be further integrated into our lives without being intrusive (a la Google Glass) – it is a certainty that augmented reality provides opportunities to enhance user experiences beyond measure.

This will almost certainly see major advances in the much-hyped but still little seen; Internet of Things.

UX designers in the AR field will need to seriously consider the questions of how traditional experiences can be improved through AR – just making your cooker capable of using computer enhancements is not enough; it needs to healthier eating or better cooked food for users to care.

The future will belong to AR when it improves task efficiency or the quality of the output of an experience for the user. This is the key challenge of the 21st century UX profession.

Conclusion:

Augmented reality is a transformative technology with a wide range of applications that extend beyond entertainment. As AR hardware and software continue to evolve, it is likely to play an increasingly significant role in our daily lives, changing the way we work, learn, and interact with the world around us.

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9921005033

$$MOSFET \text{ Linear} : I_D = \mu C_{as} \frac{W}{L} \left((V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right)$$

ADVANCEMENTS IN ORGANIC ELECTRONICS

Organic electronics have emerged as a promising field in materials science and electronics, offering a range of unique properties and potential applications. This paper provides an overview of recent advancements in organic electronics, focusing on materials, device structures, and applications.

Introduction to Organic Electronics:

Organic electronics is a rapidly growing field that involves the development and use of organic materials in electronic devices. Unlike traditional electronics, which rely on inorganic materials like silicon, organic electronics use carbon-based materials that are flexible, lightweight, and can be produced using low-cost printing techniques. This makes organic electronics an attractive option for a wide range of applications, from flexible displays to solar cells and sensors.

A critical aspect of organic electronics is the selection of suitable organic materials. The paper discusses various organic semiconductors, dielectrics, and conductors, including organic small molecules, polymers, and hybrid materials. It also covers recent innovations in materials synthesis and processing techniques.

Organic electronic devices are diverse, ranging from organic photovoltaics (OPVs) and organic light-emitting diodes (OLEDs) to organic field-effect transistors (OFETs) and sensors. This section reviews novel device architectures and manufacturing methods, such as solution processing, printing, and roll-to-roll fabrication, emphasizing their potential for large-scale production.



Applications of Organic Electronics:

Solar Cells

- Organic solar cells can be used to generate electricity from sunlight.
- They can be made into flexible and lightweight panels for portable devices and wearables.

Lighting

- Organic light emitting diodes (OLEDs) can be used for energy-efficient lighting and displays.
- They can be made into thin, flexible, and transparent panels for various applications.

Electronics

- Organic field effect transistors (OFETs) can be used for flexible and low-power electronics.
- They can be made into thin and lightweight circuits for sensors, displays, and memory devices.



Future of Organic Electronics

Organic electronics have the potential to revolutionize the display industry with flexible and foldable screens that can be integrated into clothing or other materials. Organic electronics can be used in smart packaging to monitor and control product freshness, temperature, and other parameters. Organic electronics can be used to create low-cost, flexible, and biocompatible sensors for monitoring various health parameters.

While organic electronics have made significant progress, several challenges remain, including stability, performance, and scalability. This section outlines ongoing research efforts and future directions, such as the development of stable encapsulation techniques and the integration of organic and inorganic materials for improved performance.

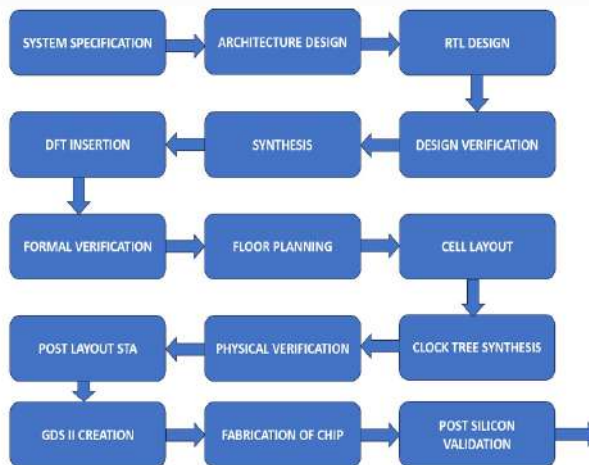
**P. Vamsi Krishna,
9922005258**

$$\text{Mass Active Law : } np = n_i^2$$

IC DESIGN FLOW

WHAT IS ASIC ?

ASIC or Application Specific Integrated Circuits can be thought of as an IC which is specifically designed to carry out certain task. Signal Processing Unit or AI/ML IC are examples for ASIC, whereas CPU cannot be considered as ASIC as can perform all the operations.



SYSTEM SPECIFICATION:

System specification is the first step in the design process of an integrated circuit (IC). It's a high-level representation of the system that defines its functional requirements and external interfaces. Factors considered in system specification include: Performance, Functionality, Physical dimensions, Interface. IC designers map out the functional requirements, verification testbenches, and testing methodologies for the project. They then turn the preliminary design into a system-level specification that can be simulated with simple models using languages like C++ and MATLAB and emulation tools. The specification of an IC design is a set of requirements that should be met across all possible operating requirements of process, voltage, and temperature.

ARCHITECTURE DESIGN:

Architecture design is a fundamental part of the integrated circuit (IC) design process. In this step, the designer gathers and discusses the requirements and specifications for the design. These specifications may include: Required speed, Power consumption, Targeted cost, Time-to-market, Required blocks, Timing constraints.

The designer also specifies the logic and circuit realization, physical placement, and even the details of the individual gates and transistors.

IC design is a sub-field of electronics engineering. ICs consist of miniaturized electronic components built into an electrical network on a monolithic semiconductor. Integrated circuit technology has made it possible to produce chips with several millions of transistors.

RTL DESIGN:

Register-transfer level (RTL) design is a hardware design methodology that describes the behavior of digital circuits. RTL design is used in the logic design phase of the integrated circuit (IC) design cycle. RTL design describes the flow of data between registers, and the operations that are performed on that data as it moves through the circuit. RTL design uses a high-level hardware description language (HDL) like Verilog, VHDL, or SystemVerilog. The circuits are described as a group of registers, Boolean equations, control logic, and event sequences. RTL design is a crucial step in the VLSI design flow, which involves the creation of electronic circuits using ICs. RTL design is converted to a gate-level description of the circuit by a logic synthesis tool. The synthesis results are then used by placement and routing tools to create a physical layout.

$$\text{Diffusion Current : } J_{n(\text{diff})} = +qD_n \frac{dn(X)}{dx} (\text{electrons})$$

DESIGN VERIFICATION:

Design verification is a critical process in the semiconductor industry that ensures the correctness of an integrated circuit (IC) design before fabrication. It ensures that the design meets the specifications required of the product and operates properly. Design verification is important because it ensures that the final product functions correctly and reliably. It also plays a key role in achieving reliable and high-performance integrated circuits. Design verification is typically obtained through: Test cases, Test suites, Inspections, Analysis.

SYNTHESIS:

Synthesis is a process in IC design that converts an abstract specification of desired circuit behavior into a design implementation in terms of logic gates. Synthesis is typically performed by a computer program called a synthesis tool. The synthesis process can be optimized for the following: Speed (timing, Area, Testability (DFT, Power (DFP, Run time. The physical effects of wiring and placement in advanced technology nodes are substantial. If the delays and voltage drops induced by long wires are not considered during the synthesis phase, it becomes very difficult to achieve the power, performance, and area requirements of the chip.

DFT INSERTION:

Design for testability (DFT) is a set of IC design techniques that add testability features to hardware product designs. DFT techniques make it easier to develop and apply manufacturing tests to the designed hardware. They cover the following areas in the life cycle of a chip: Design, Test generation.

The Synopsys TestMAX DFT tool supports all essential DFT, including boundary scan, scan chains, core wrapping, test points, and compression.

FORMAL VERIFICATION:

Formal verification is a technique that uses mathematical methods to prove the correctness of hardware designs. It can help avoid costly errors, improve reliability, and increase efficiency in hardware design.

Formal verification is an automatic checking methodology that covers all input scenarios and detects corner case bugs. It uses static analysis based on mathematical transformations to determine the correctness of hardware or software behavior. Formal verification complements traditional simulation methods. It can uncover hard-to-find corner-case bugs that elude simulation.

FLOOR PLANNING:

Floor planning is a step in VLSI design that determines the location, shape, and size of modules in a chip. It's a crucial step that decides the layout of the design.

Floor planning involves:

Placing the IC's major functional blocks, or macros
Estimating the chip area, delay, and wiring congestion

Avoiding congestion

Placing I/O pads and macros

Placing power and ground structure

Partitioning the chip area into axis aligned rectangles

Removing unnecessary placement and routing blockages

A well-optimized floor plan can lead to an ASIC design with higher performance and optimum area. Floor planning is computationally an NP hard problem. It can be challenging because it deals with the placement of I/O pads and macros, as well as power and ground structure.

CELL LAYOUT:

In integrated circuit (IC) design, cell layout is the representation of an integrated circuit in terms of planar geometric shapes. These shapes correspond to the patterns of metal, oxide, or semiconductor layers that make up the components of the integrated circuit.

Standard cells are well defined and pre-characterized cells used in ASIC (Application Specific Integrated Circuit) Design flow as basic building blocks. They have a constant size in at least one dimension that allows them to be lined up in rows on the integrated circuit.

$$\text{Mobility} : \mu_n = -\frac{\langle V_x \rangle}{E_x} (cm^2 / -sec)$$

CLOCK TREE SYNTHESIS:

Clock tree synthesis (CTS) is a technique that distributes the clock equally among all sequential parts of a VLSI design. CTS uses inverters and buffers to connect the clocks to all clock pins of sequential circuits. The goal of CTS is to balance the skew and minimize the insertion delay.

Before CTS, all clock pins are driven by a single clock source. The starting point of CTS is the clock source and the ending point is the clock pins of sequential cells.

PHYSICAL VERIFICATION:

Physical verification is a process that uses EDA software tools to ensure that an integrated circuit (IC) layout design is manufacturable and functions correctly. The process involves checking for compliance with design rules, such as:- Spacing, Metal layers, Alignment, Via placement, Short circuits, Open/float pins.

The physical verification process includes the following steps:

- Extraction
- Comparison
- Design rule checking (DRC)
- Layout-versus-schematic (LVS)
- Electrical rule check (ERC)

The DRC step determines if a chip layout satisfies a number of rules as defined by the semiconductor manufacturer. The ERC step checks a design for all electrical connections that are considered dangerous. Some tools that can be used for physical verification include:

- Calibre Design for Manufacturing
- Physical Verification System (PVS)

POST LAYOUT STA:

Static timing analysis (STA) is a method of evaluating a design's timing performance. Post-layout STA is performed after every major step in the design cycle. In post-layout STA, the net RC parameter is extracted to calculate the wire delay. The load of the cell's output is also calculated using the RC. Post-layout STA is more accurate than pre-layout STA. The STA tool analyzes all paths from each start point to each end point and compares it against the constraint (timing specification) that exists for that path.

GDS II CREATION:

GDS II (Graphics Database System or Graphic Design system) is a binary file format that's used to exchange data about integrated circuit (IC) layout artwork. It was developed by Calma in 1978. GDS II files are usually the final output of the IC design cycle. They're handed over to IC foundries for IC fabrication. GDS II files use a hierarchical structure to represent layout data. They contain information about: Planar geometric shapes, Text labels, Layers, Forms, Other geometric 2D and 3D layout data.

GDS II files have a 32 bit fixed data space, which isn't sufficient for modern IC design. GDS II files were originally written on magnetic tape.

FABRICATION OF CHIP:

The fabrication process for an integrated circuit (IC) or chip involves multiple steps:

- Wafer Preparation
- Oxidation
- Diffusion
- Ion Implantation
- Chemical-Vapor Deposition
- Photolithography
- Metallization
- Packaging

POST SILICON VALIDATION:

Post-silicon validation is the final step in the integrated circuit (IC) development cycle. It's used to detect and fix bugs in integrated circuits and systems after manufacture. The goal of post-silicon validation is to ensure that the silicon design works properly under actual operating conditions while executing real software.

**K. Chandra Kiran,
9922005115**

$$\text{Diode Equation : } I = I_0 (e^{v/nV_T} - 1)$$

5G TECHNOLOGY AND BEYOND

Introduction:

The world of telecommunications has witnessed remarkable advancements in recent years, with 5G technology standing out as a transformative milestone. 5G is set to revolutionize the way we communicate, work, and live. This introduction provides an overview of 5G technology, its significance, and the profound impact it's already having on our interconnected world. However, the journey does not stop here; the evolution of connectivity continues.

In this article, we delve into 5G technology and explore what lies beyond, including 6G and the future of wireless communication. It offers significantly faster data speeds, lower latency, and increased network capacity compared to its predecessor, 4G. This technology's implications span various sectors, from healthcare to transportation, entertainment to manufacturing.

5G Technology: A Game-Changer

- **Speed:** 5G provides blazing-fast speeds, enabling downloads, uploads, and streaming at unprecedented rates. This is crucial for applications like augmented reality (AR), virtual reality (VR), and high-definition video conferencing.
- **Low Latency:** With latency as low as one millisecond, 5G allows real-time interactions, which is critical for autonomous vehicles, remote surgery, and online gaming.
- **Massive IoT:** 5G can connect a massive number of devices simultaneously, facilitating the growth of the Internet of Things (IoT) and smart cities.
- **Enhanced Reliability:** The technology ensures a more stable and dependable connection, making it suitable for mission-critical applications.
- **Energy Efficiency:** 5G networks are designed to be more energy-efficient, reducing the carbon footprint of wireless communication.
- **Network Slicing:** 5G introduces the concept of network slicing, which allows a single physical network infrastructure to be divided into multiple virtual networks with different characteristics. This enables customized services for various industries and applications.

- **Edge Computing:** 5G complements edge computing, where data processing occurs closer to the data source. This reduces latency and enhances the performance of applications like augmented reality and autonomous vehicles.



Impact on Industries:

- **Healthcare:** 5G enables remote surgery and real-time patient monitoring, revolutionizing healthcare delivery.
- **Manufacturing:** With 5G, factories can embrace automation and predictive maintenance, increasing efficiency and reducing downtime.
- **Transportation:** Autonomous vehicles and smart traffic management systems rely on 5G for instantaneous communication.
- **Entertainment:** Virtual reality, augmented reality, and 4K/8K streaming are all made more immersive and accessible thanks to 5G.
- **Smart Cities:** 5G infrastructure supports smart city initiatives, optimizing resource management and enhancing urban living.



Beyond 5G: What's Next?

6G Technology:

6G is the next generation of wireless technology currently under research and development. While 6G is in its infancy and not expected to be widely available until the 2030s, it promises to push the boundaries of wireless communication even further. Some potential features include:

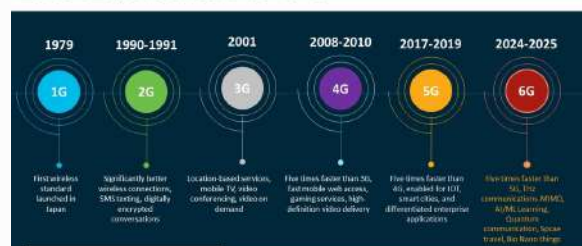
- **Speed Beyond Imagination:** 6G is expected to offer data speeds that are hundreds of times faster than 5G, potentially reaching terabits per second. This would enable instantaneous downloads of entire movies in a fraction of a second.
- **Ubiquitous Connectivity:** 6G aims to provide seamless connectivity everywhere on Earth, including remote and underserved areas. It may involve satellite networks and stratospheric balloons.
- **AI Integration:** Artificial intelligence (AI) will play a pivotal role in 6G, optimizing network management, improving security, and enabling advanced applications.
- **Holographic Communication:** 6G could introduce holographic telepresence, allowing people to interact as if they were in the same room, regardless of their physical location.
- **Environmental Sustainability:** Environmental concerns are likely to be addressed, with a focus on reducing energy consumption and e-waste.
- **Augmented Reality (AR) and Virtual Reality (VR):** 5G and beyond will enable more immersive AR and VR experiences, from interactive gaming to professional applications like architectural design and medical training.



Challenges Ahead:

- **Technical Complexity:** Developing 6G will be technically challenging, requiring breakthroughs in materials, antennas, and signal processing.
- **Security Concerns:** As networks become more sophisticated, they also become more susceptible to cyber threats. Ensuring robust security will be paramount.
- **Regulatory and Ethical Issues:** The deployment of 6G will raise questions about privacy, data ownership, and regulatory frameworks.
- **Affordability and Accessibility:** Making advanced technology accessible to all, including underserved communities, is a considerable challenge.

The Evolution to 6G



Conclusion:

5G technology and its future iterations, often referred to as "beyond 5G" or "6G," represent a remarkable evolution in the world of wireless communication. These advancements hold the promise of transforming the way we connect, communicate, and interact with the digital world. However, achieving these goals will require collaboration between industry, academia, and policymakers to overcome technical, regulatory, and ethical challenges.

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NEUROMORPHIC COMPUTING

Introduction:

Neuromorphic computing is a cutting-edge field of computer science and engineering that draws inspiration from the intricate workings of the human brain to revolutionize the way we process information and perform computational tasks. Unlike traditional computing models, which rely on digital bits and binary logic, neuromorphic computing seeks to emulate the parallel and energy-efficient processing observed in biological neural networks.

In this fascinating realm of technology, researchers and engineers are developing hardware and software systems that mimic the fundamental principles of neural processing. By doing so, they aim to create machines that can learn, adapt, and perform complex tasks with remarkable efficiency and speed, much like the human brain.

This introduction will explore the fundamental concepts behind neuromorphic computing, its potential applications, and the exciting promise it holds for the future of artificial intelligence, robotics, and beyond. As we delve deeper into this captivating field, we will uncover the mechanisms driving neuromorphic computation and its potential to reshape the landscape of intelligent systems.

Neuromorphic computing represents a paradigm shift in the world of artificial intelligence and computer science. It is a multidisciplinary field that seeks to design computing systems inspired by the intricate architecture and functioning of the human brain. This revolutionary approach aims to overcome the limitations of traditional computing models by harnessing the principles of neurobiology to build machines that can process information with unprecedented efficiency, adaptability, and intelligence.

Neuromorphic computing seeks to emulate the efficient and adaptive information processing capabilities of the human brain using hardware and software systems. However, several challenges exist in the field of neuromorphic computing that need to be addressed to fully realize its potential. Addressing these challenges in neuromorphic computing will pave the way for the development of more efficient, versatile, and ethically responsible brain-inspired computing systems that can revolutionize various fields, from artificial intelligence to neurobiology.

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Abstract:

Neuromorphic computing is a field of research and technology that aims to design and build computer systems inspired by the structure and function of the human brain. The term "neuromorphic" is derived from "neuro" (relating to neurons) and "morphic" (meaning shape or form), signifying the effort to mimic the neural structure and processes of the brain in the architecture of these computing systems. We define neuromorphic computers as non-von Neumann computers whose structure and function are inspired by brains and that are composed of neurons and synapses. Von Neumann computers are composed of separate CPUs and memory units, where data and instructions are stored in the latter. In a neuromorphic computer, on the other hand, both processing and memory are governed by the neurons and the synapses. Programs in neuromorphic computers are defined by the structure of the neural network and its parameters, rather than by explicit instructions as in a von Neumann computer. In addition, while von Neumann computers encode information as numerical values represented by binary values, neuromorphic computers receive spikes as input, where the associated time at which they occur, their magnitude and their shape can be used to encode numerical information.

Neuromorphic Computing in AI:

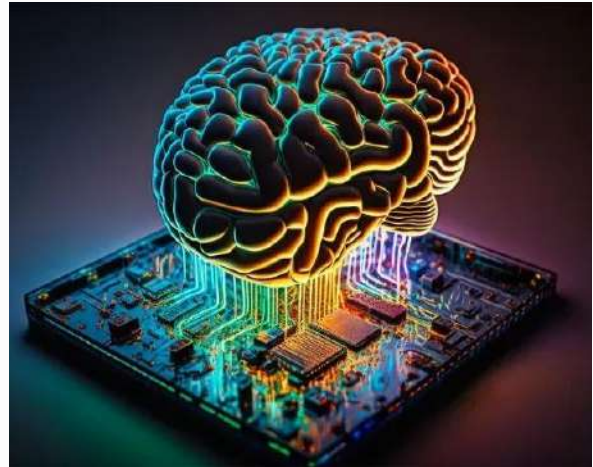
Neuromorphic computing is a crucial technology for the development of artificial intelligence systems. Based on the structure and function of biological neurons, neuromorphic computing systems can achieve a level of flexibility and adaptability unmatched by traditional AI algorithms.

Neuromorphic computing can learn and adapt in real-time. Compared to traditional AI algorithms, which require significant amounts of data to be trained on before they can be effective, neuromorphic computing systems can learn and adapt on the fly. This means they can quickly respond to changing environments and situations, making them ideal for use in robotics and self-driving cars.

Neuromorphic computing is also energy-efficient. By using electronic circuits and devices inspired by the structure and function of biological neurons, neuromorphic computing systems can potentially use less power than traditional AI systems. This could have important implications for developing AI systems deployed in resource-constrained environments, such as remote locations or space exploration missions. Most importantly, neuromorphic computing has the potential to unlock new capabilities for AI systems that were previously thought to be beyond their reach. For example, neuromorphic computing systems could be used to create AI systems capable of true creativity and artistic expression or that can learn and adapt in ways similar to human cognition.

Neuromorphic computing is also vital for developing AI systems that are more resilient to noise and errors. In neuromorphic computing systems, information is processed by distributed networks of neurons that work together to overcome errors and noise. They are, therefore, ideal for use in applications such as speech recognition, where noisy environments can make traditional AI algorithms less effective.

Moreover, neuromorphic computing provides privacy and security advantages. It is possible to design neuromorphic computing systems to process data locally on the device, unlike traditional AI algorithms, which rely on centralised data processing and storage. For example, biometric information and personal health data can be kept securely and privately.



Research's Done:

Neuromorphic computing research has its roots in the mid-20th century, with early efforts focused on simulating neural networks and exploring the computational principles inspired by the brain. Here are some key milestones in the history of neuromorphic computing:

McCulloch-Pitts Neuron Model (1943): The foundational work of Warren McCulloch and Walter Pitts in 1943 laid the groundwork for the concept of artificial neurons, which are fundamental components of neuromorphic computing. They developed a simplified mathematical model of a neuron that demonstrated how logical operations could be performed using networks of interconnected neurons.

Perceptron (1957): Frank Rosenblatt's development of the perceptron in 1957 marked a significant step in the early history of neural networks. The perceptron was a simplified model of a single-layer neural network capable of learning and making binary classifications.

Backpropagation Algorithm (1970s): The backpropagation algorithm, which is fundamental to training neural networks, was introduced by Paul Werbos in the 1970s. It allowed neural networks to learn and adapt their internal parameters, making them more capable of solving complex problems.

Early Neuromorphic Hardware (1980s-1990s): During the 1980s and 1990s, researchers began to develop hardware implementations of neural networks that aimed to mimic the parallel processing capabilities of the brain. These early neuromorphic systems often used custom analog circuits.

SpiNNaker Project (2006): The SpiNNaker (Spiking Neural Network Architecture) project, initiated in 2006, aimed to build a large-scale neuromorphic computing platform. It employed digital processors to simulate spiking neural networks, emphasizing the importance of parallelism.

BrainScaleS Project (2011): The BrainScaleS project, launched in 2011, focused on developing brain-inspired computing systems that operated in the analog domain. It aimed to replicate the spiking behavior of neurons more closely.

TrueNorth (2014): IBM's TrueNorth neuromorphic chip, introduced in 2014, represented a notable advancement in the field. It featured one million programmable neurons and aimed to provide energy-efficient, brain-inspired computation.

Continued Growth (2010s-Present): Neuromorphic computing research has gained significant momentum in the 2010s and continues to advance rapidly. Many academic institutions and companies are actively working on neuromorphic hardware, software, and applications. Projects like Intel's Loihi, BrainChip's Akida, and the European Human Brain Project are contributing to the field's growth.

Future Scope:

The future of neuromorphic computing is dynamic, with ongoing research and development expected to drive innovation and expand its applications across multiple industries. As technology continues to advance, neuromorphic computing has the potential to revolutionize the way we process information, interact with machines, and understand the complexities of the human brain. In the future Neuromorphic computing is going to be implemented in a maximum number of fields like Neuromorphic Hardware in Healthcare, Security and Privacy, Education, Neuromorphic Chips for Learning, Aerospace, Entertainment, Gaming etc... Thus Neuromorphic computing is going to rule in all these fields, at present, we can see some of the machines, cars, robot's ect.. based on AI, we can say that the most evolving technology is the Neuromorphic Computing.

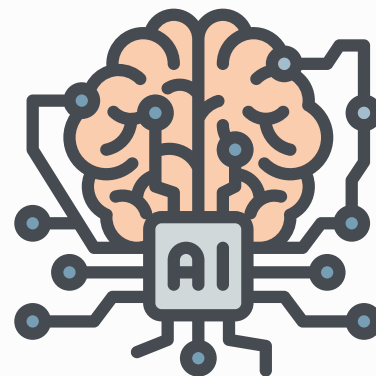
Neuromorphic computing is poised to make significant strides in various industries in the future. From healthcare to education, security to entertainment, the integration of neuromorphic hardware is set to bring about groundbreaking changes. Picture a world where technology is seamlessly integrated into our daily lives, enhancing how we learn, work, and play.

As we witness the evolution of artificial intelligence in machines, cars, and robots today, it is evident that the next frontier lies in the realm of neuromorphic computing. Its adaptability and potential for innovation make it a frontrunner in shaping the technological landscape of tomorrow. Get ready to witness the transformative power of neuromorphic computing across diverse fields - the possibilities are endless!

Conclusion:

Neuromorphic computing represents a remarkable convergence of biology and technology. Its results are poised to transform industries ranging from AI and robotics to healthcare and neuroscience research. With its unparalleled energy efficiency, real-time processing capabilities, and potential for cognitive computing, neuromorphic computing is set to shape the future of technology in ways we are only beginning to comprehend.

As research and development continue to progress, the full extent of its impact remains to be seen, but one thing is clear: neuromorphic computing is a force to be reckoned with on the frontier of technological innovation. Neuromorphic computing holds immense promise for revolutionizing the way we process information, perform AI tasks, and understand the human brain. While significant challenges remain, ongoing research and development in this field have the potential to unlock groundbreaking applications across various domains, from AI and robotics to neuroscience and beyond. As the technology matures, it will be exciting to see how neuromorphic computing contributes to the advancement of science and technology.



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THE RESISTOR'S REBELLION

Once upon a time, in the vibrant city of Electronica, a bustling metropolis that ran on the principles of electronics, there was a delicate balance maintained by electronic components. This balance was crucial to keep the city running smoothly. In this city, transistors acted as gatekeepers, allowing or blocking the flow of data, and capacitors stored and released energy like diligent caretakers.

The story began with a power supply unit named Volar, which was the heart of Electronica. Volar was responsible for providing energy to all the electronic components in the city. It had been serving the city dutifully for years, tirelessly converting alternating current (AC) into direct current (DC) and distributing it to the necessary components. However, Volar was ageing, and its energy output was becoming less stable.

One fateful day, Volar's energy fluctuated uncontrollably, causing chaos throughout the city. The capacitors, who were responsible for smoothing out these fluctuations, were overwhelmed. They tried their best but needed help. They called upon their friend, Resistor to have the unique ability to control the current flow and resist the surges. Resistor had an idea. It knew that the transistors, which were typically shy and introverted components, had the potential to assist. It sent out signals to the transistors, urging them to act as amplifiers to boost the capacitor's efforts. The transistors were initially hesitant, but Resistor convinced them that this was their moment to shine. Together, the capacitors and transistors are allied, creating a network of stabilizers throughout the city. This network began regulating the energy fluctuations and restoring order. Meanwhile, Inductor, a wise inductor living on the outskirts of Electronica, sensed the imbalance and sent out electromagnetic waves to aid in the process.

As the capacitors, transistors, and inductor worked in harmony, Electronica gradually returned to its vibrant self. The city's insulators, who had the important role of keeping current where it belonged, applauded the efforts of their fellow components.

However, a new challenge emerged when a malicious diode named Dr. Zener started causing disturbances, trying to let unwanted currents flow into sensitive areas of the city.

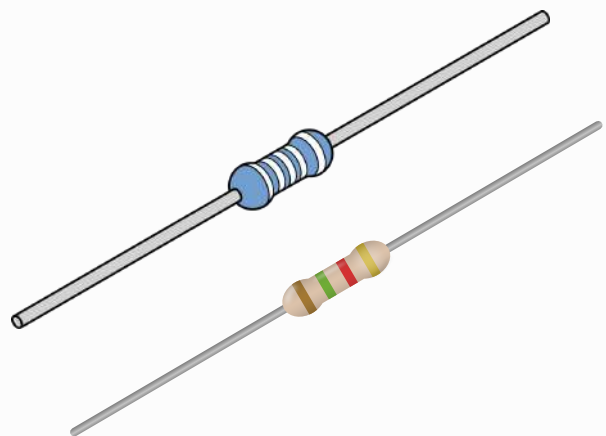
The diode was cunning and persistent, but the vigilant transistors, with their newfound confidence, joined forces with the diodes on the opposite side to block Dr. Zener's sinister plans.

The battle between the diodes and the transistors raged on, but the diodes' attempts were futile against the determined transistors, who used their amplifying powers to outwit Dr.

Zener at every turn. With the help of the transistors and their alliance with the capacitors, Electronica remained safe from unwanted intrusions.

CONCLUSION :

In the end, harmony was restored to Electronica, thanks to the collaboration of its electronic components. Volar, the power supply unit, realized the importance of adapting and working together with its fellow components to ensure the city's stability. The story of Volar, Resistor, Capacitors, Transistors, Inductor, and the valiant Insulators became a legend, a reminder that unity and cooperation were the keys to a thriving Electronica.



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QUANTUM DIMENSIONS

Once graphics, there objects in an upon was a time in the world of computers a need to represent three-dimensional two-dimensional space, which to the development of various techniques and algorithms to handle dimensions effectively.

One of the curliest challenges was the concept of ers and mathematicians worked dili Projection. Eng Engineers diligently to devise methods for projecting 30 objects onto a 20 plane, a porous known as “dimension reduction. Was essential for creating realistic images in video games and simulations.



In the 1960s, Ivan Sutherland introduced the Concept of “3D computer graphics and developed the first computer program called ‘sketchpad, which Could manipulate objects in three dimensions. This marked computing. A significant milestone in dimension in Fast forward to the 1980s, and the field of computer graphics witnessed the birth of 3D modelling and rendering Software like Autodesk’s AutoCAD and Pixar’s RenderMan. These tools allowed engineers and artists to design and visualize objects in a 3D space, which was then transformed into animations. 20 images es or In the ever-evolving world of technology, the story of dimensions continues to unfold, from si graphic high dimensional data to high Putting, the understanding remains analysis and quantum com- analysis and manipulation of dimensions central to many technological advancements.



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INSPIRING PERSONALITIES



When it comes to tech-related content, one can hardly miss Technical Guruji by Gaurav Chaudhary. He is a social influencer specialising in creating tech-based Hindi content for his 23 million fans. One of the reasons he is often considered no. 1 vlogger in India is because of how he connects with fans and shows how Hindi can be used to create tech content. His other channel, Gaurav Chaudhary, is mainly related to his travelling and has already roped in 5.21 million members as a family. Even though he lives in Dubai, he happens to be a hardcore Indian.

Technical Guruji, is a YouTube channel that aims to simplify tech topics in Hindi. Gaurav Chaudhary is driven by the mission of making technology accessible to everyone, ensuring that anyone with an interest in tech can easily grasp it. Technical Guruji consistently delivers two videos daily, covering the latest in technology and tech news. For those seeking Hindi-language content to stay updated in the tech world, subscribing to Technical Guruji is a smart choice.

TechBurner by Jaskirat Singh simplifies complex tech concepts in a formal, educational manner for viewers interested in staying informed about technology trends. The channel caters to a wide audience, offering structured explorations of technology realms, making intricate details easy to grasp and keeping viewers enlightened and inspired. TechBurner serves as a beacon of knowledge in the ever-evolving world of technology, providing valuable insights and a clear understanding of the latest innovations.

With a formal and educational approach, the channel ensures that viewers are not only informed but also engaged in the latest technology trends. Whether you're a newbie looking to expand your tech knowledge or a seasoned tech lover seeking deeper insights, TechBurner offers a structured journey through the vast realms of technology, making even the most complex details accessible and exciting. By serving as a trusted source of knowledge, TechBurner shines a light on the ever-evolving landscape of technology, empowering viewers with valuable insights and a clear understanding of the cutting-edge innovations shaping our world today. So, sit back, relax, and let TechBurner guide you through the fascinating world of tech with clarity and inspiration.



Become Keyboard Master With These 20 Useful Computer Keyboard Shortcut Keys With a mission to educate and enlighten, Technology Gyan, hosted by Manoj Saru is a trusted resource for tech enthusiasts seeking comprehensive insights. The channel maintains a professional and methodical approach in its content, making it a reliable destination for individuals looking to deepen their understanding of the tech world.



Technology Gyan is now one of the most well-known technology-related channels in India. Manoj Saru aka Technology Gyan an Indian tech YouTuber and Blogger was born in 1994 in New Delhi. Manoj has made it to the index of Famous Indian YouTubers and is known as Top Indian Tech YouTuber. Currently Manoj has Over 96 lakh Subscribers on his YouTube Channel and 4.5 lakh + Followers on Instagram Handle (@manoj_saru), 1.8 Million Followers on Facebook & 2 lakh + Followers on Twitter. Manoj Saru was a center kid in his examinations since his school days and was not as capable as different understudies, however his insight into method was far superior to his school days, which made him a Successful person today.

Code With Harry is all about making coding easy. Harry, the creator, teaches coding in a straightforward way, and he shares code and resources on his website, CodeWithHarry.com. You can also find the code in the video descriptions. The best part about this channel is that Harry also teaches programming in Hindi, so it's accessible to a wide audience.

Code With Harry is truly a gem for anyone looking to dive into the world of coding. With Harry at the helm, learning to code becomes an enjoyable and enriching experience. His dedication to simplifying complex coding concepts and providing valuable resources sets him apart in the realm of online coding tutorials. By offering his content in Hindi, Harry ensures that aspiring programmers from diverse backgrounds have the opportunity to enhance their skills and knowledge. So whether you're a beginner looking to start your coding journey or an experienced coder seeking to refine your craft, Code With Harry is the perfect platform to learn, grow, and succeed in the exciting field of programming.





FIND THE WORD?

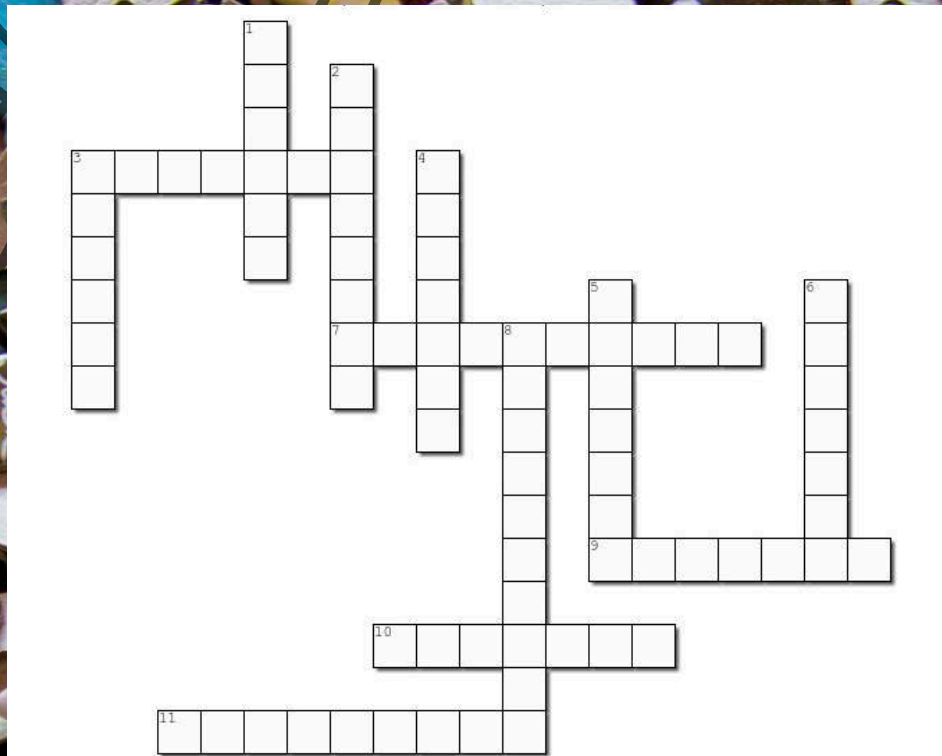
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C	A	T	W	C	E	A	D	T	M	R	Q	R	S	A
B	E	J	R	E	B	L	D	I	I	T	A	N	E	I
R	J	O	Z	G	I	S	T	L	M	C	A	K	S	H
T	F	G	F	E	O	Y	V	D	A	C	E	M	A	M
E	K	X	C	I	N	O	S	A	R	T	L	U	Q	E

ACCELEROMETER
SENSORS
VIBRATION
LIGHT

FORCE
PRESSURE
TEMPERATURE

HUMIDITY
PROXIMITY
ULTRASONIC

CROSSWORD-1



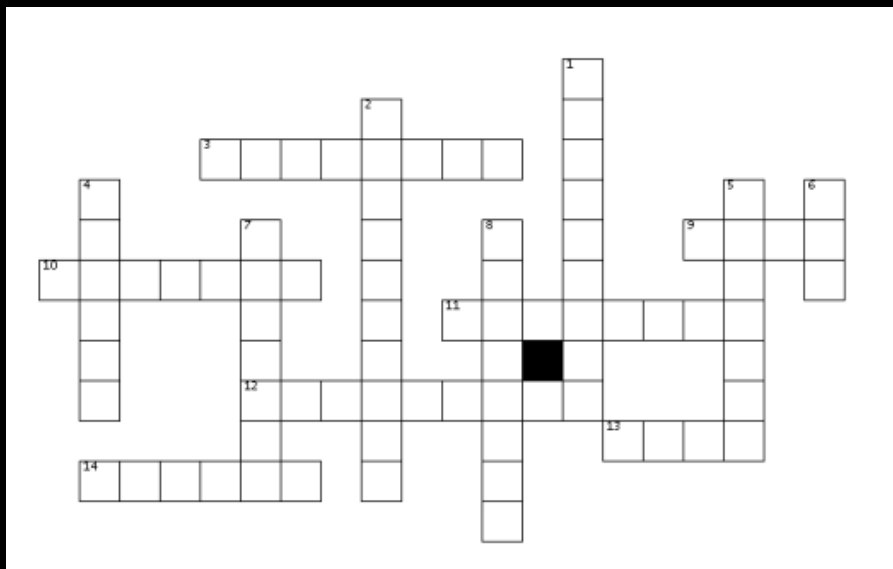
Across:

- 3. Device used to measure current
- 7. Current Considered to movements of
- 9. A substance found only in it's pure form
- 10. Voltage source. Which converts chemical to electrical Energy
- 11. Material which opposes flow of electricity

Down:

- 1. Opens or completes an Electric Path
- 2. A Voltmeter is used in which circuit
- 3. One coulmb passing a point in one second (unit of Current)
- 4. A flow of Electric Charge
- 5. Term designate Electrical Pressure
- 6. Part of an atom has no Electric charge
- 8. Semiconductor Device used to amplify or to switch Electrical Signal

CROSSWORD-2



ACROSS:

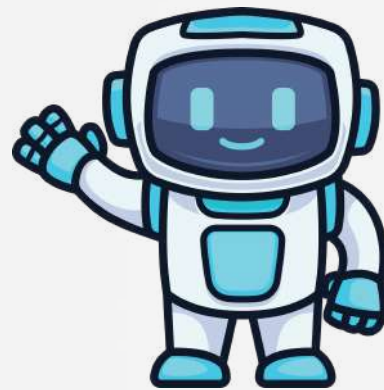
3. IEEE 802.3 standard.
9. A small electrical component designed to protect electronic appliances from excessive flow of current
10. It is a closed loop or a pathway through which electric current flows.
11. _____ is used to restrict the flow of current in a circuit.
12. The component that stores electrical energy in the form of electrical charge is_____
13. SI unit of power is_____
14. What is the 7th pin of IC7408?

DOWN:

1. A electronic device used to take an input signal and produce an output signal with a higher or lower intensity
2. I'm not for eating, but I help you connect
4. What is the name of the electrical component with ON & OFF state?
5. Movement of electrons produces _____
6. I don't need much power, yet I shine so bright, who am I?
7. A semiconductor material with voltage drop of 0.7 is _____
8. A negative dancer around nucleus.

AI BOTS

- ChatGPT-Solves Anything
- Copy.AI- Writes Anything
- Mid Journey-Create Art
- Tome.app-Create PPTs
- Soundraw.io-Create Music
- Pictory- Script to Video
- Firefiles-Notetaking
- Runway-Edit Videos
- Krisp- Removes Background voices, noises, and echo
- Ocoya-Create and Schedule social media content 10x faster
- Unbounce Smart Copy-Write high performing Cold emails at scale
- Quickchat- AI chatbots that automate customer service chats

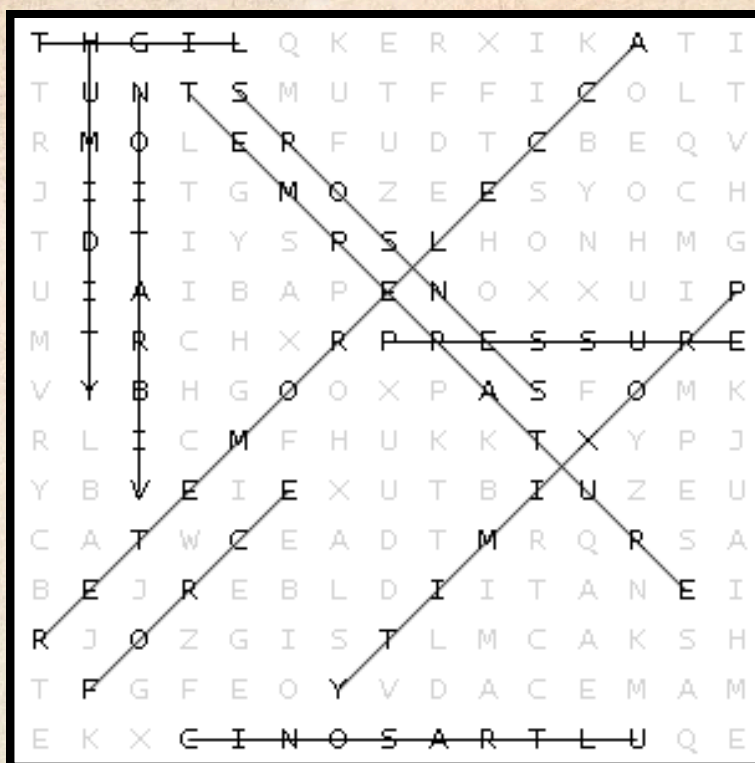


USEFUL WEBSITES

- NPTEL - An online learning platform that offers free engineering and science courses.
- IEEE Xplore - A digital library of engineering and technology research articles, conference proceedings, and technical standards.
- edX - An online learning platform offering courses and degree programs from top universities worldwide.
- One Compiler - A free online compiler for writing, running, and sharing code in over 60 programming languages.
- Instructables - A website providing step-by-step instructions for DIY projects, including electronics and 3D printing.
- TED Talks - A website featuring informative and inspiring talks by experts in various fields.
- Free Code Camp - A comprehensive curriculum for learning web development and data science.
- Futurepedia - A library of AI-based tools for a wide range of applications.
- Grammarly - A writing assistant that helps improve grammar, spelling, and style.
- Gutenberg - A library of over 60,000 free e-books that can be downloaded or read online.
- Tools120 - A website offering free online conversion and calculation tools.
- Tinkercad - A free web app for 3D design, electronics, and coding.
- GitHub - A web-based platform for managing and sharing code with the world's largest coding community.
- Arduino - An open-source electronics platform that enables users to create interactive electronic objects.
- Canva - a website that helps students create professional-looking designs and graphics for presentations, posters, and social media.



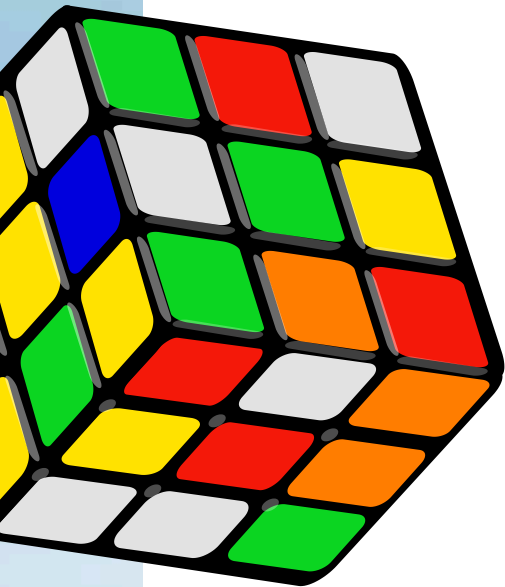
FIND THE WORD ANSWERS



c

o

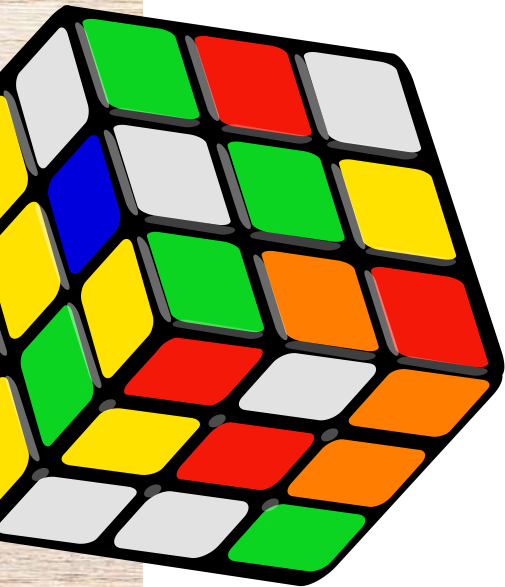
- 1) SWITCH
- 2) PARALLEL
- 3) AMPERE
- 3) AMMETER (ACROSS)
- 4) CURRENT
- 5) VOLTAGE
- 6) NEUTRON
- 7) ELECTRONS
- 8) TRANSISTOR
- 9) ELEMENT
- 10) BATTERY
- 11) INSULATOR



C

D

- 1)AMPLIFIER
- 2)BREADBOARD
- 3)ETHERNET
- 4)SWITCH
- 5)CURRENT
- 6)LED
- 7)SILICON
- 8)ELECTRON
- 9)FUSE
- 10)CIRCUIT
- 11)RESISTOR
- 12)CAPACITOR
- 13)WATT
- 14)GROUND





Essays

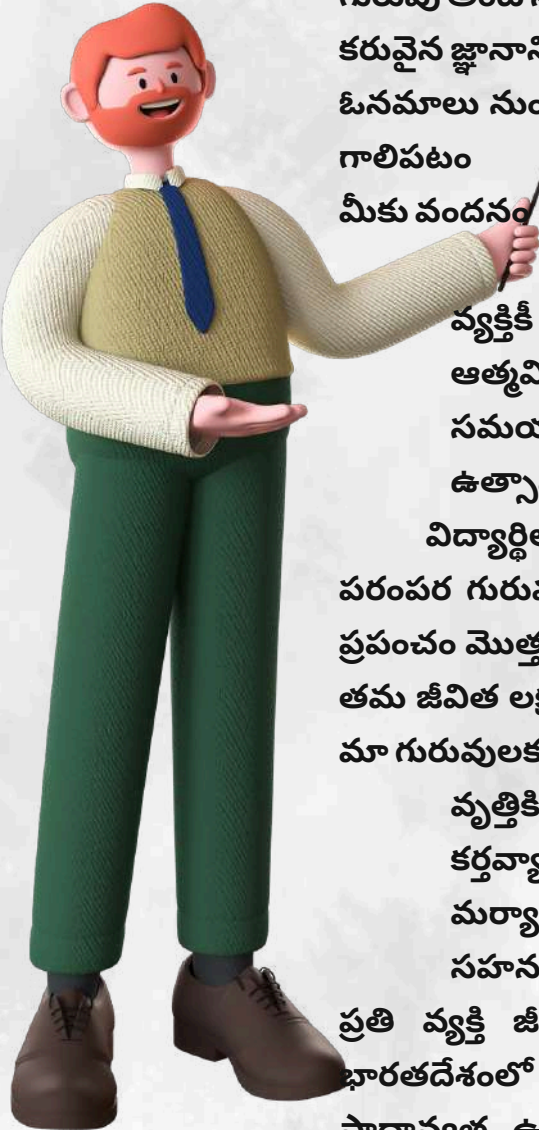


గురువు గొప్పదనం



అజ్ఞానాధికారంలో జన్మించిన నాకు జ్ఞానదీపంతో కళ్ళు తెరిపించిన గురువులకు నా గౌరవపూర్వక ప్రణామములు.

సమాజంలో జరిగే మంచి ఏదైతే మనకు కనిపిస్తుందో ఆ మంచిని నేర్పేది, ఆ మంచికి పునాది గురువు. అనాదిగా వస్తున్న ఆచారాలను సంస్కృతి సంప్రదాయాలను మన నరనరాల్లోకి ఎక్కించే ఒకే ఒక్క ఆయుధం గురువు. గురువు అంటే సమాజానికి ఒక గౌరవం. గురువు అంటే మన గెలుపుకు ఎరువు. కరువైన జ్ఞానానికి బరువైన పదసంపదను వరంగా ఇచ్చేది సరస్వతి. బడిలోని ఓనమాలు నుండి మన అమ్మఒడి గుడి గోప్పతనాలను వివరించేది గురువు. గాలిపటం విద్యార్థి అయితే దానికి ఆధారమైన దారం గురువు. “అందుకే మీకు వందనం అభివందనం.”

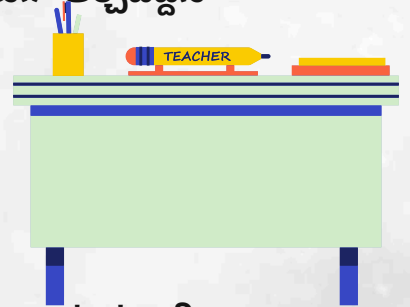


వ్యక్తికి వ్యక్తిత్వంతో వన్నె తెచ్చిన వివేకి
ఆత్మవిశ్వాసమే శ్వాసగా పయనించే విహారి
సమయానికి సంస్కారం నేర్పని సాహసి
ఉత్సాహానికి ఊపిరి స్నేహానికి సారథి లాంటి వారు గురువులు...



విద్యార్థిలోని అజ్ఞానాన్ని తొలగించేవారు గురువు అందుకే భారతీయ పరంపర గురువులకు ఓ గొప్ప స్థానాన్ని కల్పించింది. ఆ స్థానాన్ని ప్రస్తుతం ప్రపంచం మొత్తం నిలుపుకోవాల్సిని అవసరం ఎంతైనా ఉంది. ఆ బాధ్యతను తమ జీవిత లక్ష్యంగా చేసుకొని మనలందరిని ఉత్తమ పౌరులుగా తీర్చిదిద్దిన మా గురువులకు వందనం, అభివందనం, పాదాభివందనం.

వృత్తికి వివేకమద్దిన వినయశీలి
కర్తవ్యానికి క్రమశిక్షణ జోడించిన సంస్కారి
మర్యాదకు మన్నన నేర్పిన నిగర్వి
సహన సౌశీల్య సమేతలు మన ఉపాధ్యాయులు...



ప్రతి వ్యక్తి జీవితంలో గురువు పాత్ర ఎంతో గణనీయంగా ఉంటుంది. భారతదేశంలో ఆధ్యాత్మికంగాను, సామాజికంగాను గురువులకు చాలా ప్రాధాన్యత ఉంటుంది. తల్లిదండ్రులు తరువాత గురువు అంతడివాడిని మాతృదేవోభవ, పితృదేవోభవ, ఆచార్యదేవోభవ. అనే సూక్తి చెబుతుంది.



ఒక తల్లి లేదా తండ్రి కేవలం తమ కుటుంబాలపైన మాత్రమే ప్రభావం చూపవచ్చు, కానీ ఒక గురువు సమాజం పైనే తీవ్ర ప్రభావాన్ని చూపవచ్చు. వినయంలేకుండా అహంకారం ఉన్నచోట ఎంత చదువు చెప్పిన అది సమాజ బ్రష్టత్వానికే కారణం అవుతుంది. నిత్య కృషివలుడు ఎవరు అంటే అది ఉపాధ్యాయుడే.

A.P.J. Abdul kalam గారు "Indomitable spilt" అనే పుస్తకాన్ని రచిస్తూ ఒక మాట అన్నారు ఆయన, అది ఏమిటంటే ఈ దేశం మొత్తం మీద నేను గౌరవించే వ్యక్తి అది ఏదైనా ఉంది అంటే అది కేవలం ఉపాధ్యాయవృత్తి మాత్రమే అని

* ఎందుకు ఆయన అలా అన్నారు అంటే ఈ దేశ భవిష్య భవితవ్యం అంతా ఉపాధ్యాయులు చేతిలోనే ఉంటుంది, కాబట్టి సాధారణంగా మన జీవితంలో ముఖ్యమైన వ్యక్తులుగా పరిగణించేది మన తల్లిదండ్రులను మన బంధువులను ఇంకా మన స్నేహితులను యుగయుగాలుగా మానవజీవిలో చెక్కుచెదరని స్థానం ఏర్పరుచుకున్న గురువులు.

- 1) సాందీపని 2) బృహస్పతి 3) ఆదిశంకరాచార్య
- 4) ద్రోణాచార్య 5) వాల్మీకి 6) విశ్వమిత్ర
- 7) సర్వేపల్లి రాధాకృష్ణ 8) గౌతమ బుద్ధుడు
- 9) చాణక్యుడు 10) సావిత్రిబాయి పూలే
- 11) రవీంద్రనాథ్ ఠాగూర్ 12) అబ్దుల్ కలాం
- 13) స్వామి వివేకానంద

* వీళ్ళలో పైన చెప్పిన బంధాలు, బాంధవ్యాలు కలిగిన మనషులు ప్రప్రథమంగా మనకు కనిపిస్తారు. కానీ వీరిందరి తరువాత మన జీవితంలో చాలా ముఖ్యమైన పాత్ర పోషించే వ్యక్తులను మాత్రం మన చాలా అరుదుగా గుర్తు పెట్టుకుంటాము వారే గురువు. సృష్టిలో ఏ మనిషికైన తొలి గురువు అమ్మ. అమ్మ తరువాత అలాంటి అత్యుత్తమ స్థానాన్ని భగవంతుడు ఇంకో రూపంలో గురువుకు మాత్రమే ఇచ్చాడు. ఎప్పటికైనా మనలోని నైపుణ్యాన్ని గుర్తుపట్టే సూక్ష్మ జ్ఞానం గురువుకు మాత్రమే ఉంటుందిని మనం తెలుసుకోవాలి.





ఉపాధ్యాయులు గురువులు ఎప్పటికీ ఒక్కటి కాదు. ఎలానో ఇప్పుడు తెలుసుకోండి. పాఠాన్ని పుస్తకంలో చూసి ఆపాఠాన్ని అర్థం చేసుకొని దానిని తిరిగి పిల్లలకు బోధించడం ఉపాధ్యాయుడు. చేసే పని. కానీ గురువు చేసే పని చాలా గొప్పది అనే చెప్పాలి. తనలోని జ్ఞానాన్ని ఎళ్ళ తరబడి తాను సాధించిన సంపాదించిన అనుభవాన్ని రంగరించి పాఠంలా చెప్పగలిగితే అతనిని గురువు అని మనం సంభోదించాలి. పాఠాలనే కాకుండా గుణపాఠాలను కూడా నేర్పడం గురువు యొక్క కర్తవ్యం. చదువు చెప్పి నుంచి ఉత్తీర్ణత సాధించడంతో ఉపాధ్యాయుడి బాధ్యత తీరిపోతుంది అని అనుకుంటారు. కానీ గురువు చదువుకే కాకుండా శిష్యుల వ్యక్తిగతా మరియు మానసిక ఎదుగుదలకు సైతం తోడ్పడాలి.

కనిపించే ప్రతి వారిలో గురువు ఉంటాడు. కేవలం గురువు అన్న పీటానెక్కిన ప్రతివారు గురువుతారు అని అర్థం. అలా గురువు స్థానం చాలా గొప్పది అని తెలుసుకోవాలి.

ఇప్పుడు మన గురువులు Government Service నుండి retire అయినా గురువు అనే వృత్తి నుండి retire కాలేరు. తమ చివరి క్షణం వరకు ఎంతో మంచి మంచి పౌరులను దేశానికి అందిస్తూనే ఉంటారు.

ఉత్తమ ఉపాధ్యాయులుగా, మానవతా వాదులుగా, విద్యావైత్తగా అందరి హృదయాల్లోనూ పదిలమైన స్థానాన్ని సంపాదించుకున్న మన గురువులకు మన ఎంతో ఋణపడి ఉన్నాం.

గురుభ్యోనమః

ప్రపంచమంతా నిశ్శబ్దంగా నిద్రపోతున్న సమయంలో మీరు తిప్పే పేజీ చప్పుళ్లు.....

భవిష్యత్తులో విజయం సాదించిన తరువాత ప్రపంచం కొట్టే చప్పుళ్లు కావాలి.



P. Jyothi
9922005287
II - B ECE



Student Life



Welcome to Student life

Balancing Academics and social life

Getting involved on campus

Managing stress and mental health

Exploring career options

Making the most of your college experience



Welcome to student life:-

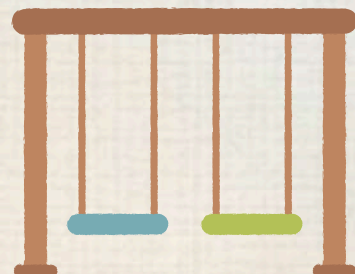
Welcome to the exciting world of Student life. As you embark on this new journey, you will discover a world of opportunities Challenges, and experiences that will shape your future in ways you never thought possible.

Balancing Academic and Social Life:-

Finding a balance between academics and social life is crucial for a successful college experience while it's important to Prioritize your studies, neglecting your social life can negatively affect Your mental health and overall well-being.

Getting involved on Campus:-

Joining a club or organisation on a great way to meet new people who share Your interests. Whether it's a sports team, a cultures group, or an academic club, there are plenty of options to choose from. By getting involved, you will not only make new friends but also gain valuable leadership and teamwork skills that will benefit you in your future career.



Managing stress and mental Health:

College can be stressful. with academic Pressure, social obligations and time, Challenges & adjusting to a new environment all taking their toll. It is important to take steps to Manage Your Stress and maintain your mental health to stay healthy and happy throughout your College career.

Exploring career options: Exploring career options early on in your College career can be a game changer. I can help you gain valuable experience mind your network, and set yourself up for success after graduation. These Opportunities not only provide hands-on experience but also give you the chance to make connections in your field.



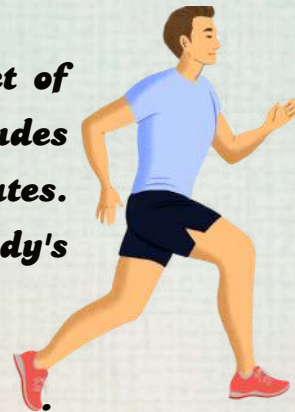
K. Praveen Kumar
9922005119

Habits for maintaining healthy lifestyle

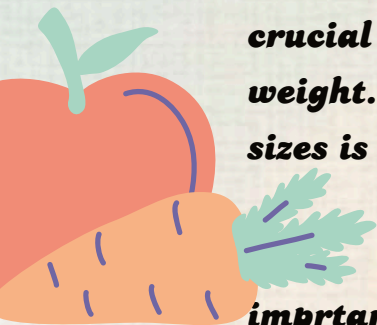


Certainly. nutrition and dietary habits for maintaining a healthy lifestyle are important non-technical. topics. Here are some key considerations in this regard:

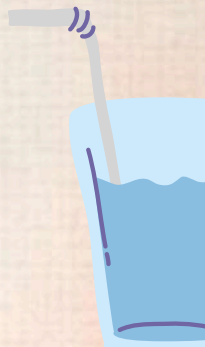
(1)- Balanced Diet:- A fundamental aspect of healthi nutrition is maintaing a balanced diet This includes Consuming appropriate Portions of carbohydrates. Proteins. fats, vitamins and minerals to meet the body's nutritional needs.



(2) Portion control: understaning portion sizes is crucial in preventing overeating and maintaing a healthi weight. Teaching individuals how to gauge approrfu Portion sizes is a non-technical. but essential skill.



(3) Hydration: Educating individuals on the imprtane of staring. adequately. hydrated is non-technical. but crucial.



(4) meal Planning Teaching individuals how to plan balanced meals and snacks can help them make. healthier food choices and avoid reliance. on. Convenience foods. which may be less nutritions.



**D.V. Madhu Sudhan Reddy
9922005029**

Nutrition and dietary Habits for Healthy

Drink fat-free and eat low-fat or low-fat milk and dairy products. Choose a variety of low-fat Sources including eggs, beans Poultry and Protein without Skin, Sea food, lean meats, and unsalted If you four nuts, you eat meat times more Seeds and Soy Products. eat white meat at least more often than red meat.

Healthy Eating Food Habits to Achieve Your Health Goals:

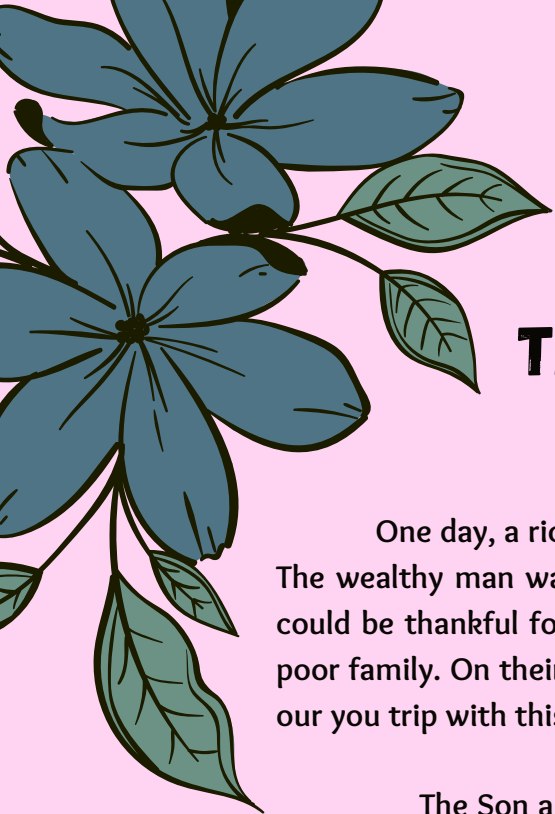
1. Eat more Home-cooked Meals.
2. Include a lot of greens.
3. Do not Skip your breakfast.
4. Add more Protein.
5. Cut down Sugar.
6. chew Properly
- 7..Keep Yourself Hydrated.
8. Precise mindful eating.
9. Go for Evenings Walks.
10. Eat more Veggies.



Importance of a healthy diet A healthy diet is a habit essential for good health and nutrition. It protects you against many chronic noncommunicable diseases, Such as heart disease, diabetes and Eating a variety of foods and Cancelling. consuming less Salt, Sugars and Saturated and Industrially essential for Producing trans-fats healthy diet.

K. Indu Reddy
9922005042





THE REAL WEALTH



One day, a rich man decided to take his son on a trip to the country. The wealthy man wanted to show his son how the poor people live so he could be thankful for his wealth. They Spent two days on a farm of a very poor family. On their own way home, the rich man asked his son "How was our you trip with this poor family and what did you learn?"

The Son answered, "I Saw we have servants to serve us, but they serve others, we have one dog and they have four. We have imported lanterns in our garden and they have stars at night. We buy oir food, but they grow theirs. we have a big pool in our garden, but they have a Creek that has a no end. We have large walls to protect us and they have friends to protect them.

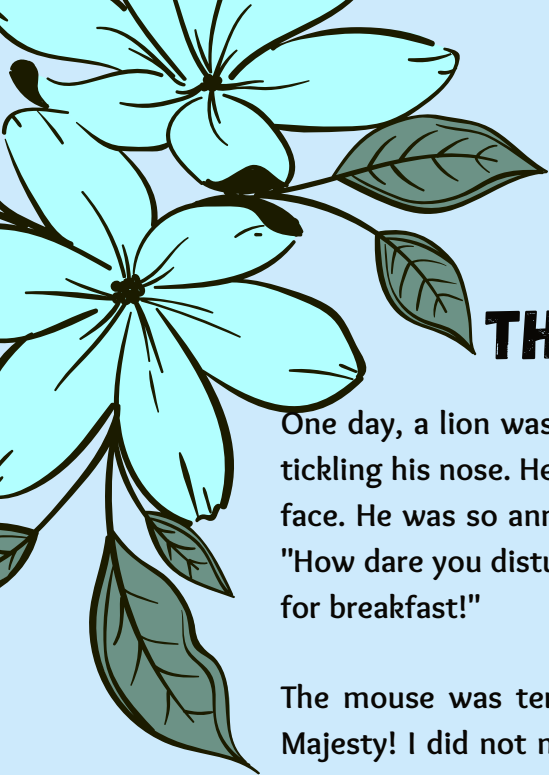
Finally, the Son added, "Thanks dad for showing me how poor we are."

Moral: True Wealth isn't Money or materials. It is friendship and Kindness.

It is friendship and Kindness



S. Vijay Kanth
9921005296



THE LION AND MOUSE

One day, a lion was sleeping peacefully in his den when he felt something tickling his nose. He opened his eyes and saw a tiny mouse running over his face. He was so annoyed that he caught the mouse in his paw and roared, "How dare you disturb my sleep, you little pest! I will crush you and eat you for breakfast!"

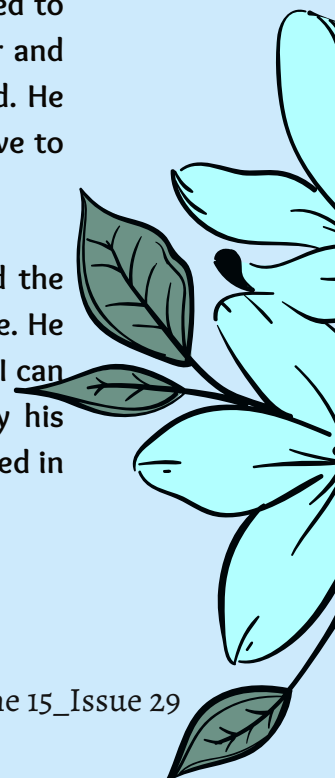
The mouse was terrified and begged for mercy. "Please, spare me, Your Majesty! I did not mean to wake you up. It was an accident. Please let me go, and I will be forever grateful. Maybe someday I can repay your kindness."

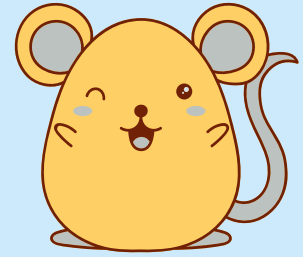
The lion laughed at the mouse's words. "You, repay me? How can a weak and tiny creature like you help a mighty and powerful king like me? You are nothing but a snack for me. But I am feeling generous today, so I will let you go. But don't ever bother me again, or you will regret it." He opened his paw and let the mouse scamper away.

The mouse thanked the lion and ran as fast as he could. He was glad to be alive and hoped to never see the lion again.

A few days later, the lion was hunting in the forest when he stepped into a hunter's trap. He felt a net fall over him and heard a loud snap. He tried to free himself, but the net was too strong and tight. He roared in anger and frustration, but no one came to help him. He realized he was doomed. He thought, "This is the end for me. I will be killed by the hunters or starve to death in this net. I wish I had someone to help me."

Meanwhile, the mouse was passing by the same area when he heard the lion's roars. He recognized the voice of the lion who had spared his life. He thought, "This is the same lion who let me go. He is in trouble. Maybe I can help him. After all, he was kind to me once. It is my duty to repay his kindness." He followed the sound of the roars and found the lion trapped in the net.





He saw the lion's sad and hopeless eyes and said, "Don't worry, Your Majesty. I am here to help you. You may not remember me, but you saved my life once. Now it is my turn to save yours. Please stay calm and quiet. I will gnaw through the net and free you."

The lion was surprised and touched by the mouse's words. He said, "You are the mouse I spared? You came back to help me? How can you do that? The net is too thick and strong for your teeth."

The mouse said, "Nothing is impossible if you have courage and determination. Trust me, Your Majesty. I will not let you down." He began to nibble at the net with his sharp teeth. He worked hard and fast, cutting through the ropes one by one. He did not stop until he made a big hole in the net.

He said, "There, Your Majesty. You are free. Please get out of the net quickly before the hunters come back."

The lion thanked the mouse and said, "You are amazing, my friend. You have done what I thought was impossible. You have saved my life. You are brave and loyal. You have proved that size does not matter. You have taught me a valuable lesson. Never judge anyone by their appearance. Everyone has something to offer. Everyone deserves respect and kindness. You are the true king of the jungle."

The mouse smiled and said, "You are welcome, Your Majesty. I am glad I could help you. You are generous and humble. You have shown me that strength is not everything. You have learned a valuable lesson. Never underestimate anyone by their abilities. Everyone has something to learn. Everyone needs help and friendship. You are the true hero of the forest."

The lion and the mouse became friends and lived happily ever after.

Moral : No act of kindness is ever wasted. No one is too big or too small to help or be helped.

A Giridhar
9922005086





Pencil Art





Naveen



Naveen



Tharun





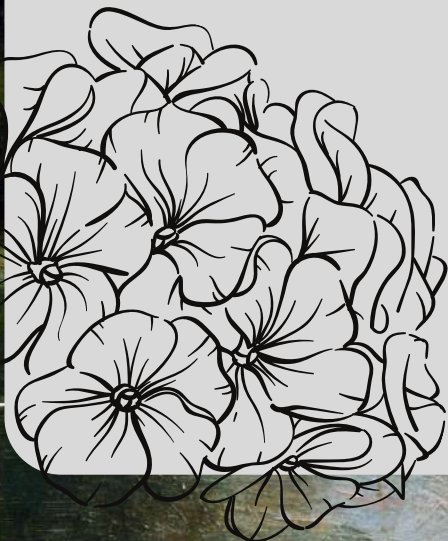
Naveen



Tharun



Naveen





Naveen



Tharun



Naveen

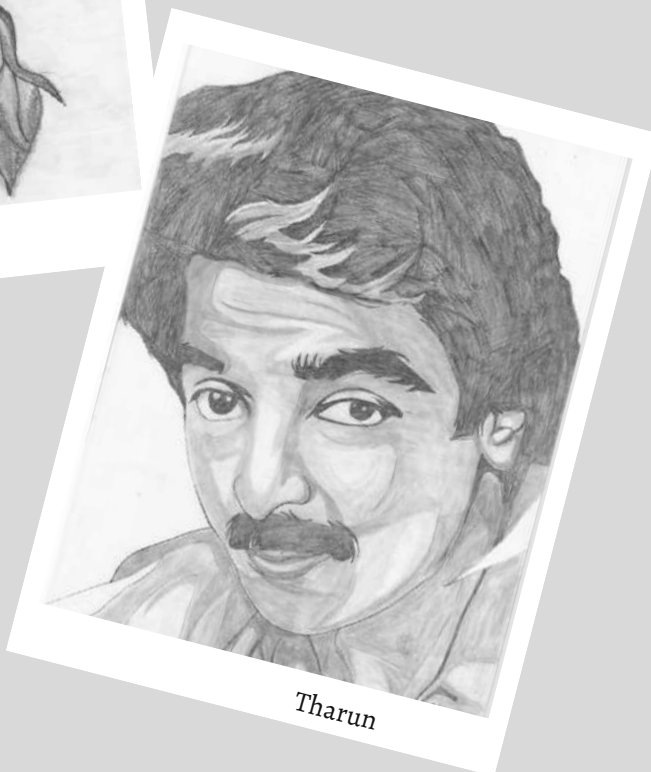




Tharun

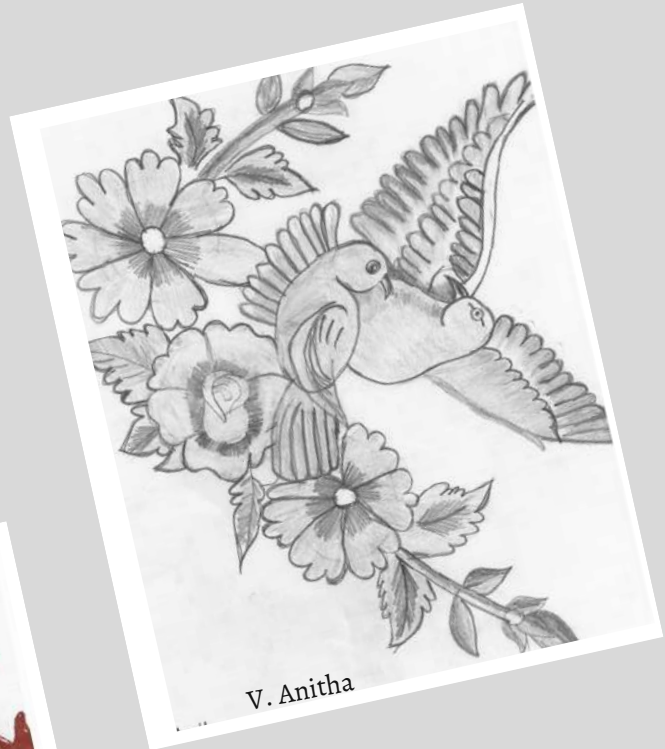


S. Aravinth - 9922005083



Tharun





V. Anitha



Tharun



Tharun





Photography



T. Srikanth



Bhoomika.U



T. Komali



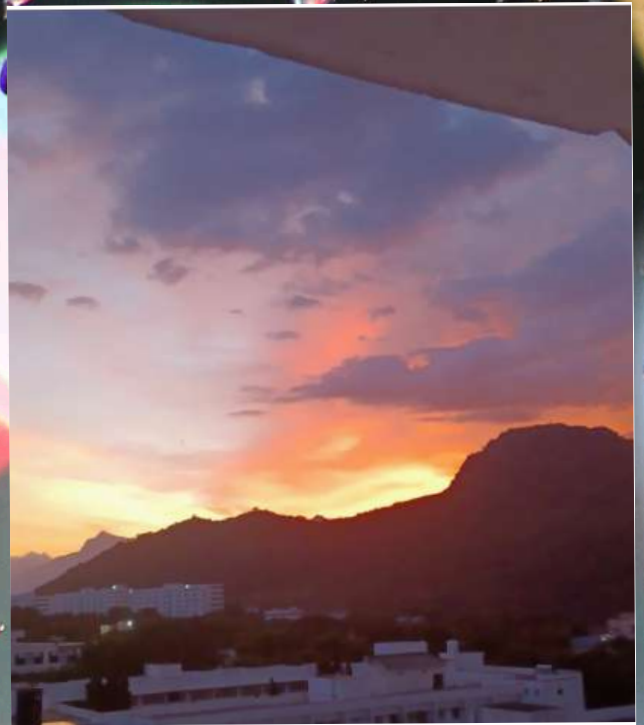
Shanmukh



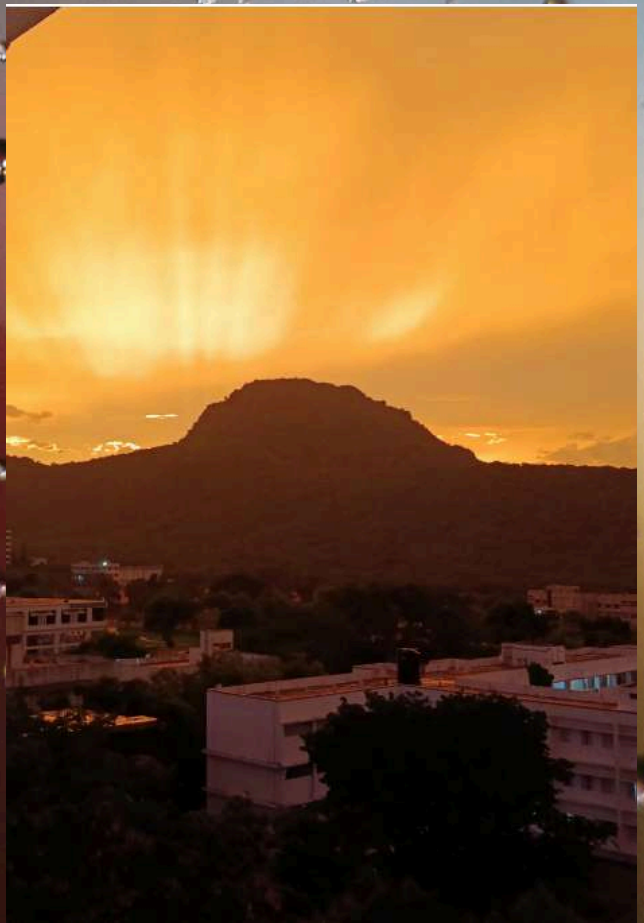
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P Vamsi Krishna



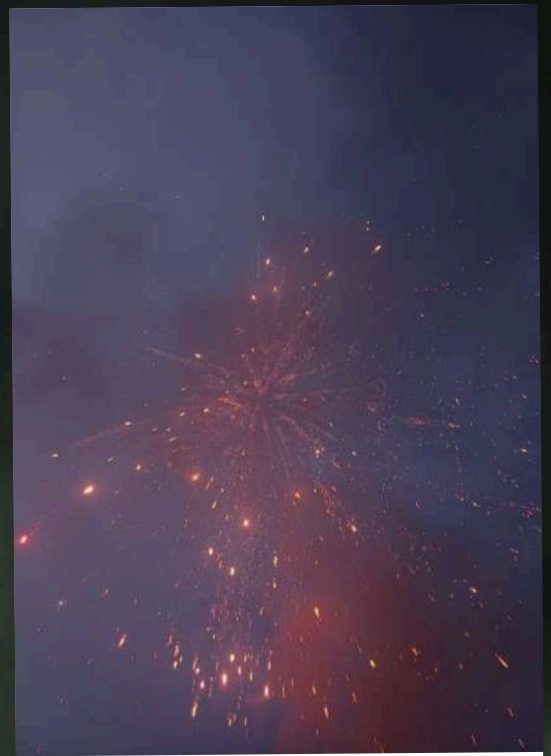
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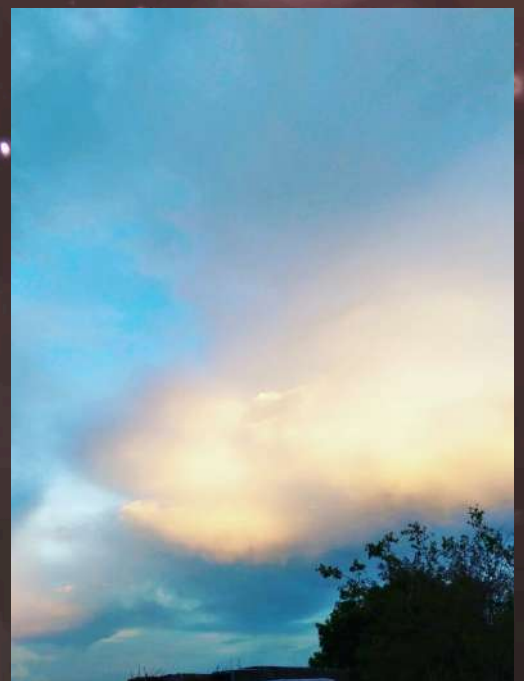
B.NarsimhaReddy



-Deepthi.G



-Deepthi.G



-Komali.T

FEEDBACK FROM OUTGOING STUDENTS



NAMBI E
Final yr., ECE
KARE

In my Department of Electronics and Communication Engineering (ECE) at KARE, my educational journey has been nothing short of exceptional. The unwavering support and friendliness of the staff have created an environment conducive to my growth and development. Beyond academics, I've had the privilege of exploring numerous opportunities to nurture both my personal and professional skills, encouraging a holistic approach to my education. With the presence of various professional societies, I've gained access to invaluable resources and networks that have significantly enhanced my learning experience. The department's proactive approach ensures that I am continually exposed to a multitude of opportunities in the field, making it the perfect place for me to thrive and excel as a budding engineer.

"College is where the pages of your life's story are written, each lecture a new chapter, each friendship a colourful character." As a student of Electronics and Communications Engineering at KARE, I am grateful for the knowledge, skills, and personal growth I have gained during my time here. The faculty's dedication to imparting knowledge is commendable, and their expertise in the field has been instrumental in shaping my understanding. The curriculum has not only equipped me with a strong foundation in the field but has also nurtured passion for innovation and technology. The curriculum is thoughtfully designed to stay current with industry trends and emerging technologies, ensuring that I am well-prepared for the ever-evolving field.



~K. VENU GOPAL
Reg No: 9921005089
III-ECE



M.Greeshma
IV/ECE-D
Reg. no: 9920005115

Students are only prepared to face the outside world in college. College is where a person's personality is developed. I could continue to feel happy that I joined KARE and the Department of Electronics and Communication Engineering. The department's main goal is to prepare students to be unique individuals who can meet the demands of the workforce. Because staff members engage closely with all students, we are able to fully understand the strengths and problems of each individual with intelligence and character. The course structure continues to learn as much as possible from practical experience while keeping up with continuous global industrial changes.

Only during their time at college are students ready to confront the outside world. Personality development occurs in college. Having joined KARE and the Department of Electronics and Communication Engineering, I could not help but feel pleased with myself. The fundamental objective of the department is to make sure that each student leaves here prepared to fulfill the expectations of the workforce. All students are closely interacted with by staff members, which allows us to comprehend each student's intelligence and character as well as their strengths and weaknesses. In keeping with ongoing developments in the global industrial landscape, the course structure continues to emphasize learning as much as possible from real-world experience.

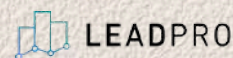
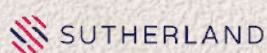


T.Sandhya
IV/ECE-D
Reg. no: 9920005069

PLACEMENTS

Our Department has an abundance of job opportunities available to students in both core and software industries, with several renowned companies like Sutherland, Synopsys, Robert Bosch, Nanchip Solutions, Tessolve Semiconductors, Renault Nissan, L&T, Accenture, TCS, HCL, Comcast, Zoho, Tech Mahindra, and Capgemini being potential employers.

Our Recruiter's



GALLERY

Our department regularly organizes workshops, events, and celebrations for a well-rounded academic experience. We are proud of our students' success in external college events and believe these provide valuable real-world opportunities for growth.

INTRAMURAL EVENTS

SPICETRON



A competition in which students participated in the circuit designing domain using the SPICE.

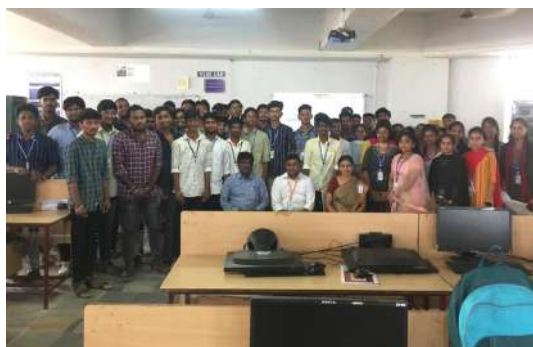
IOTRONZ



An event was conducted, where students competed with each other to showcase their projects related to the IoT domain.

INTRAMURAL EVENTS

APPDEV



An event was organized by our department where a competition was held for creating applications specifically for smart IoT devices.

BOTATHON

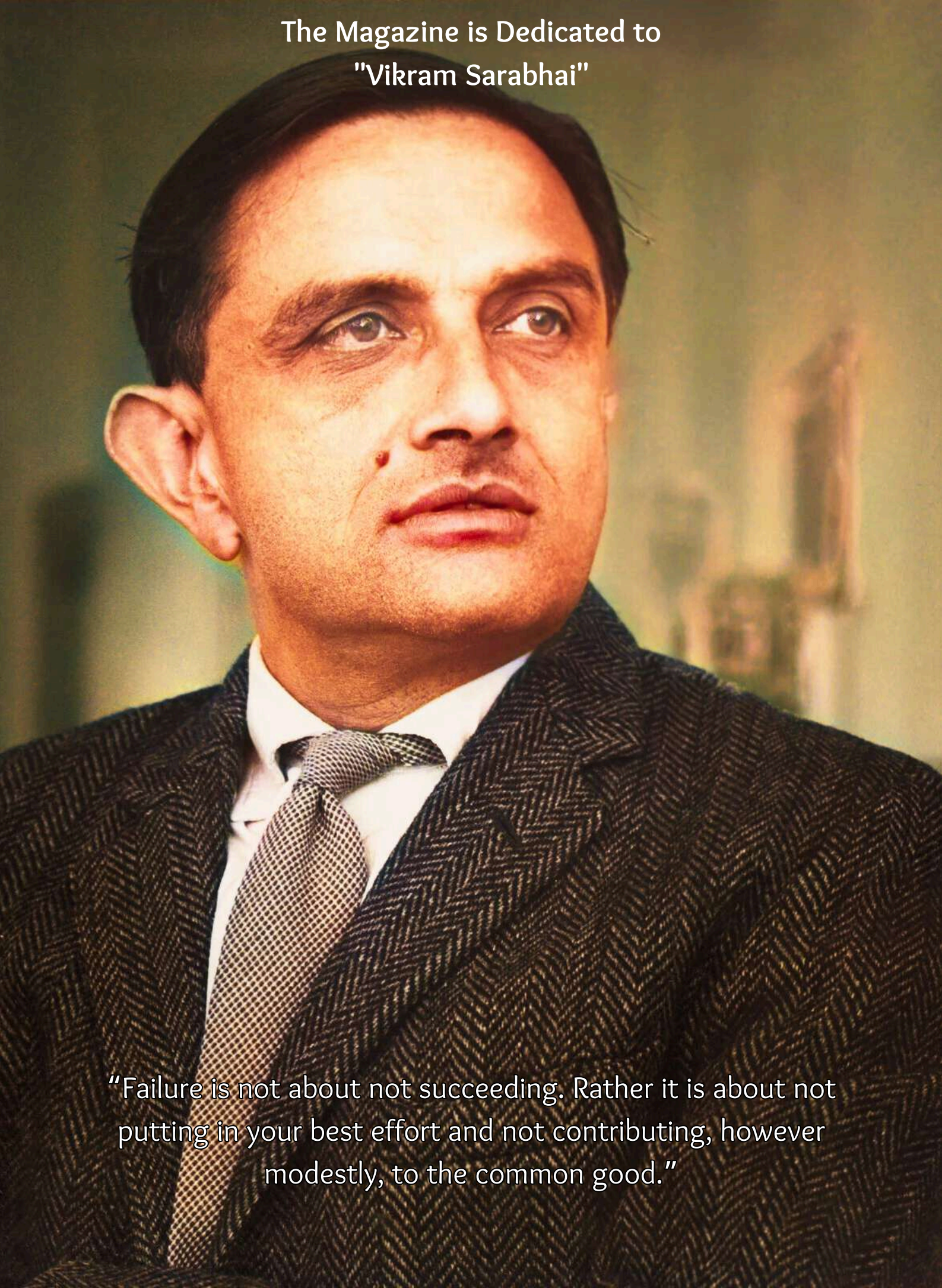


Our department organized an event where students participated in creating online interactive bots.

ARDUINON



Our department organized a hackathon competition for students based on Arduino.

A portrait of Vikram Sarabhai, a man with dark hair, looking slightly upwards and to the right. He is wearing a dark herringbone suit jacket, a white shirt, and a patterned tie. The background is a soft, out-of-focus green and yellow.

The Magazine is Dedicated to
"Vikram Sarabhai"

"Failure is not about not succeeding. Rather it is about not putting in your best effort and not contributing, however modestly, to the common good."